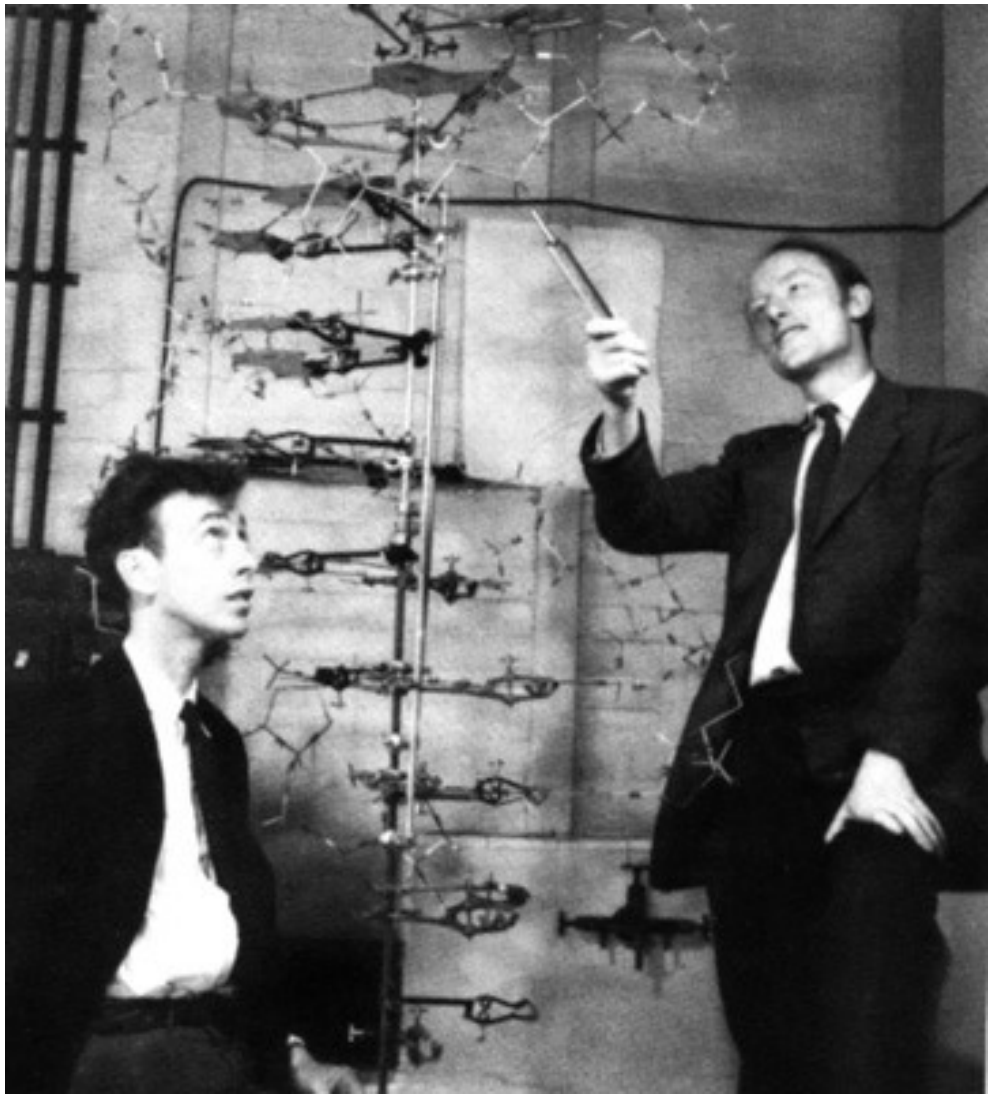
DNA



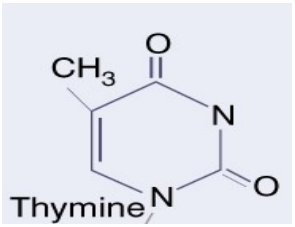


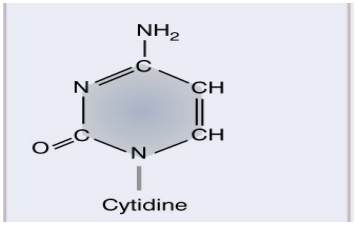
Delminda Neves FMUP

delmagal@med.up.pt

Componentes do DNA

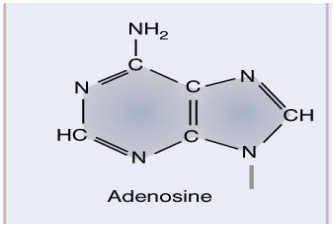
Bases Pirimídicas

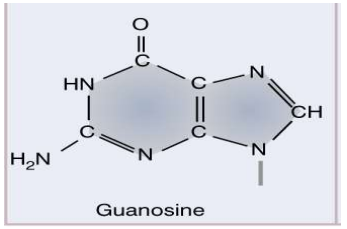




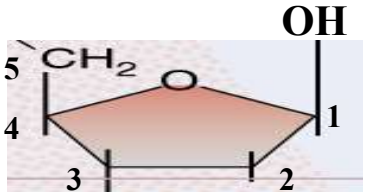
T **C**

Bases Púricas

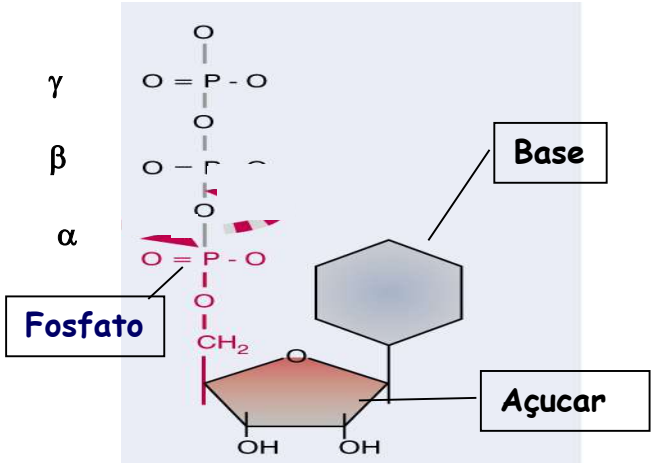




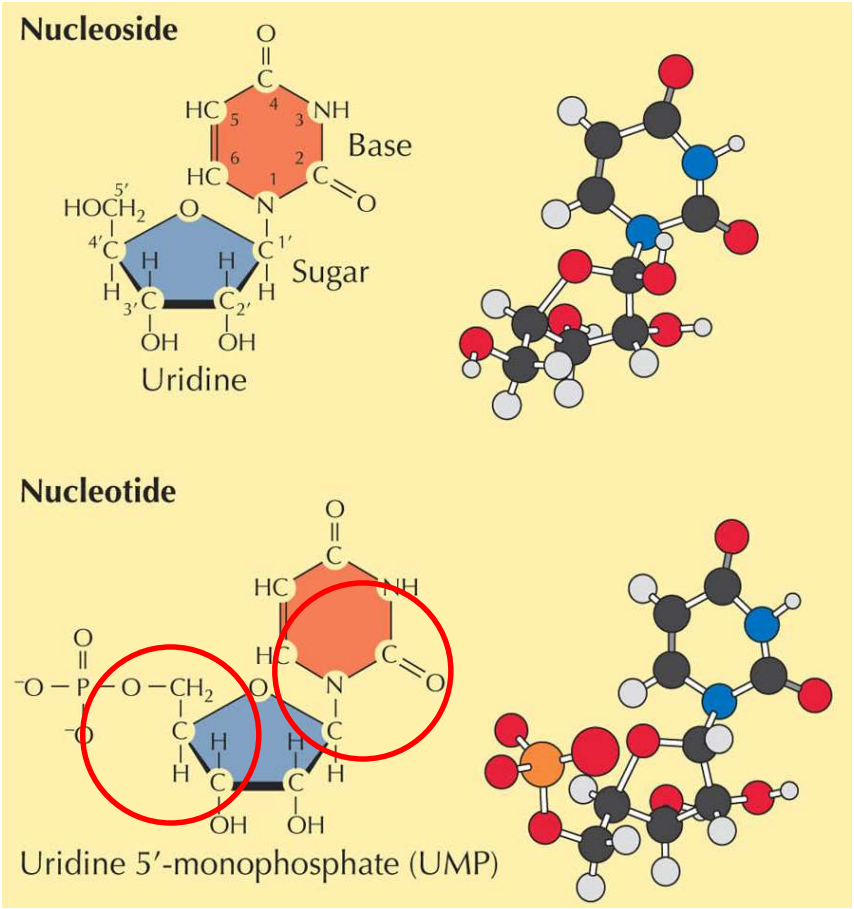
A **G**



Açucar (2-desoxirribose)



Nucleotídeo trifosfato



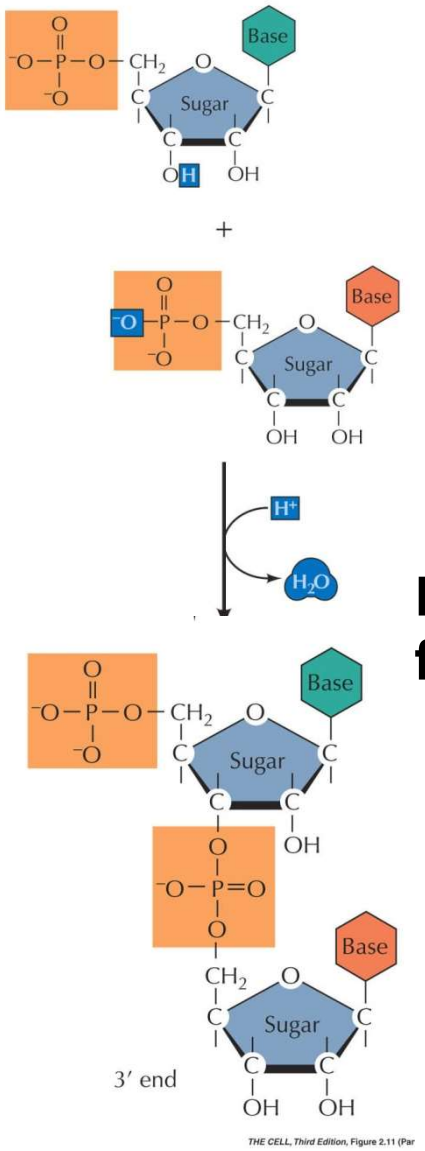
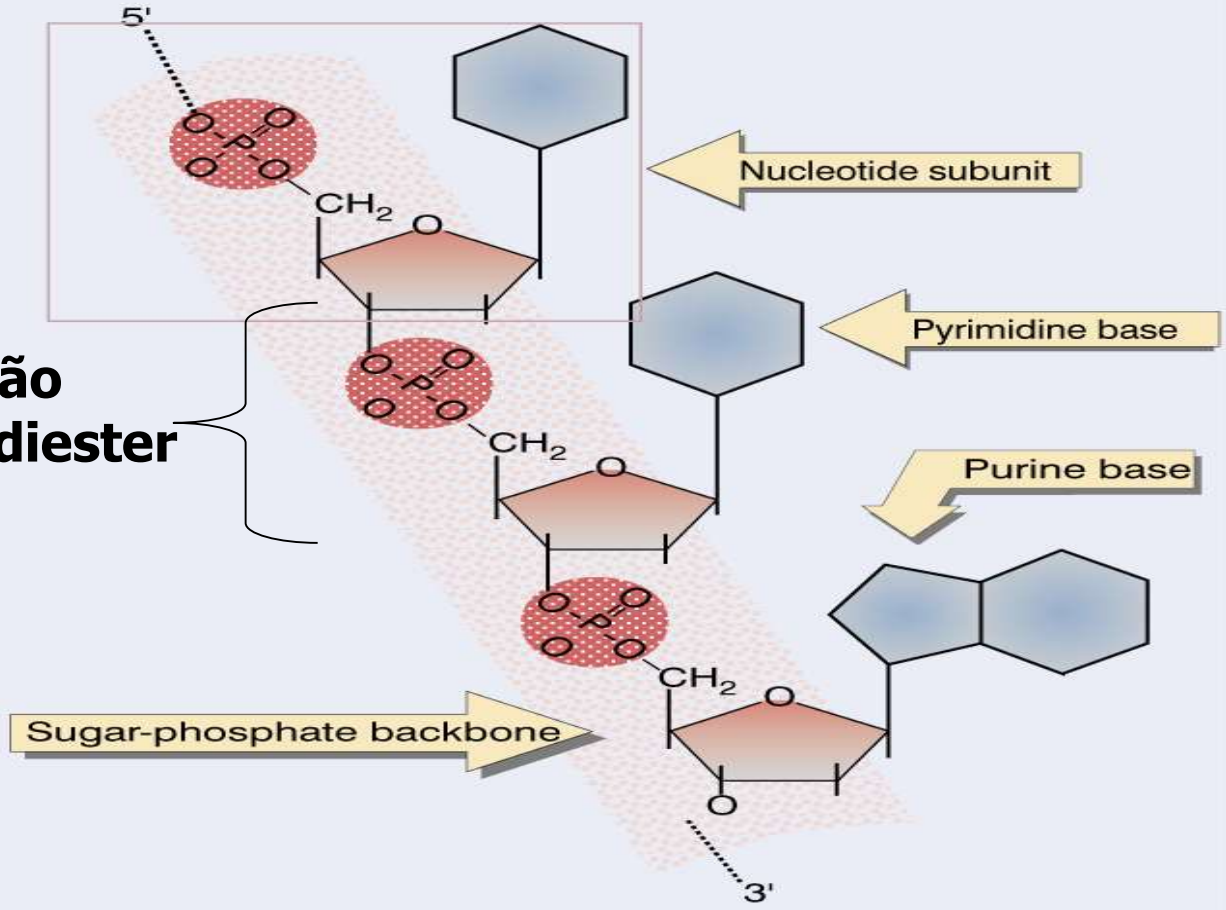


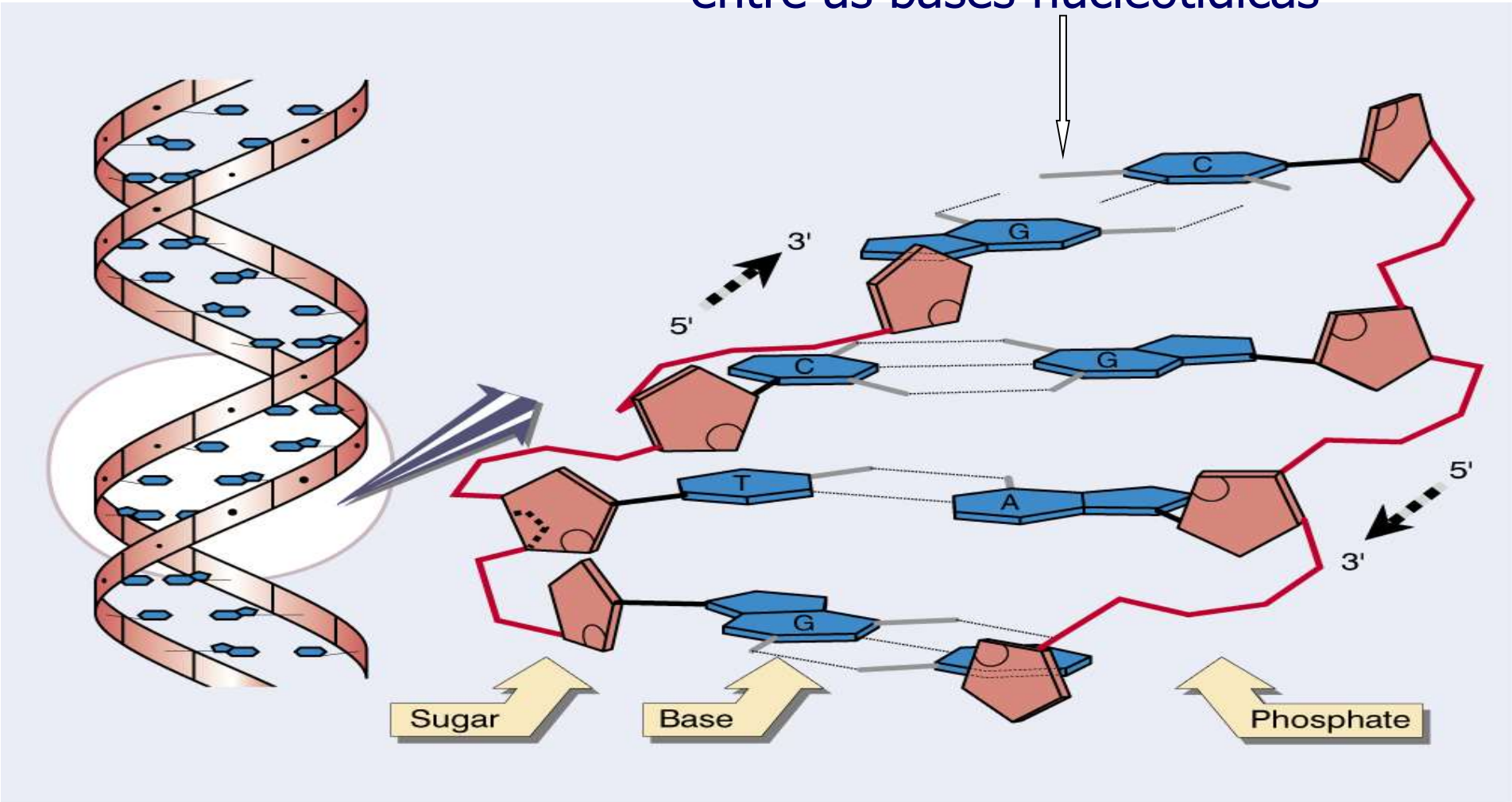
Figure 1.5 A polynucleotide chain consists of a series of 5'-3' sugar-phosphate links that form a backbone from which the bases protrude.

Ligação fosfodiester



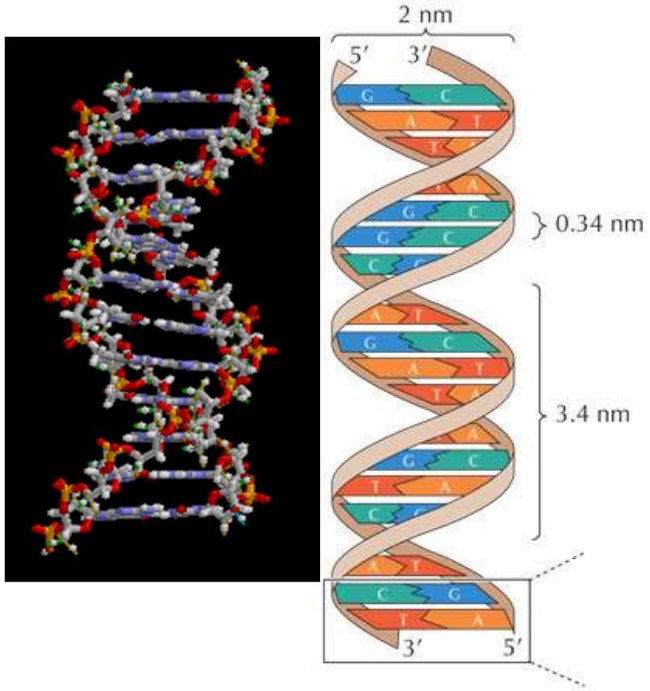
Por convenção indica-se a sequência de nucleotídeos numa molécula de DNA/RNA sempre de 5´para 3´

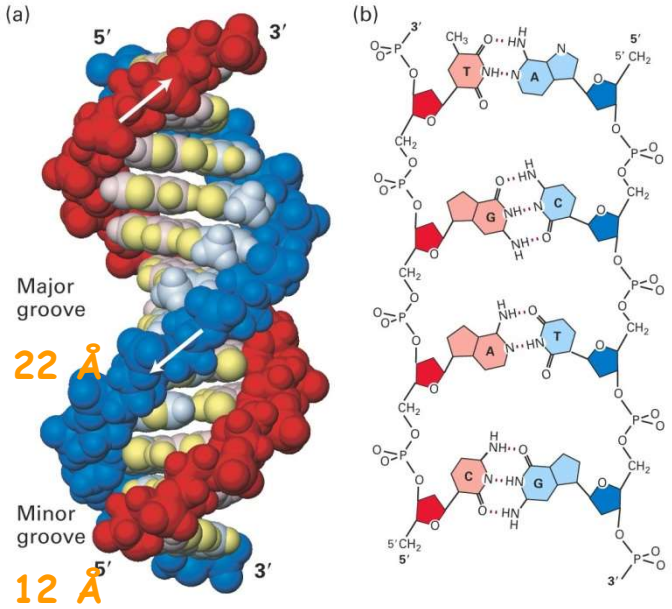
Ligações por ponte de hidrogénio entre as bases nucleotídicas

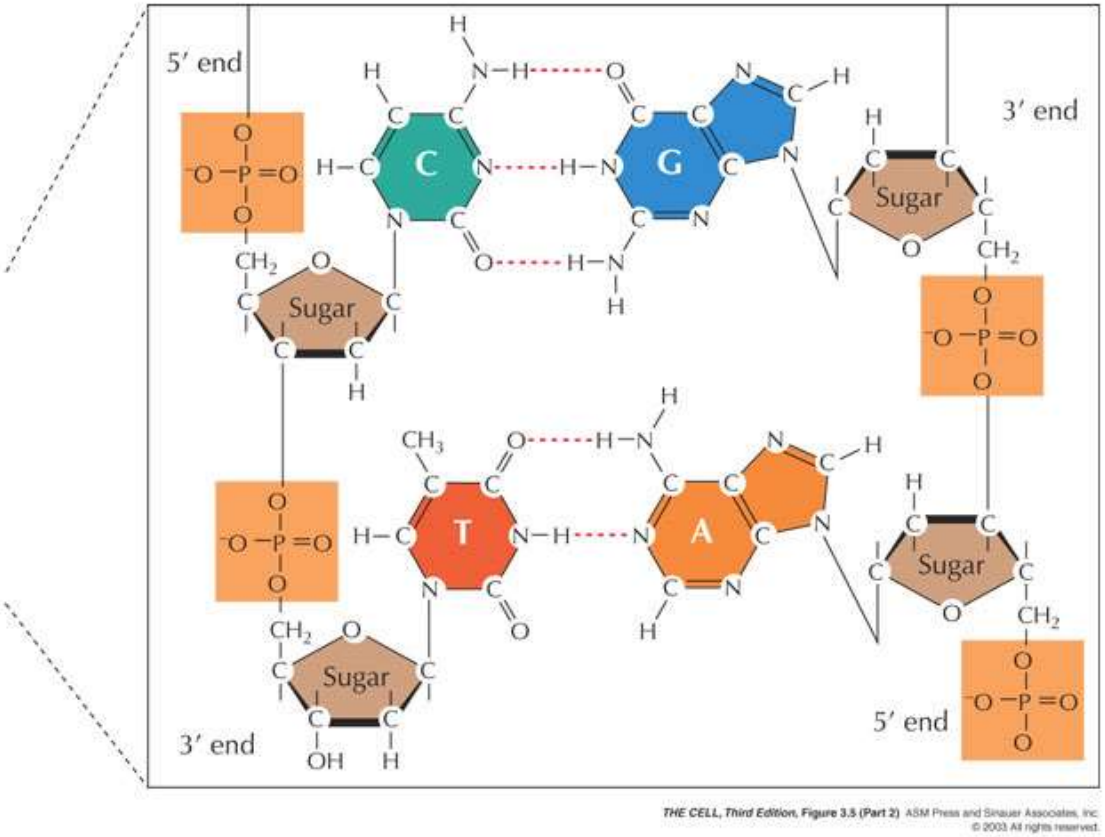


As duas cadeias açucar-fosfato ficam no exterior, enquanto as bases azotadas se projectam para o interior da dupla hélice. As bases emparelhadas dispõem-se em planos paralelos, perpendicularmente ao eixo da hélice.

Estrutura molecular do DNA







THE CELL, Third Edition, Figure 3.5 (Part 2) ASM Press and Sinauer Associates, Inc.
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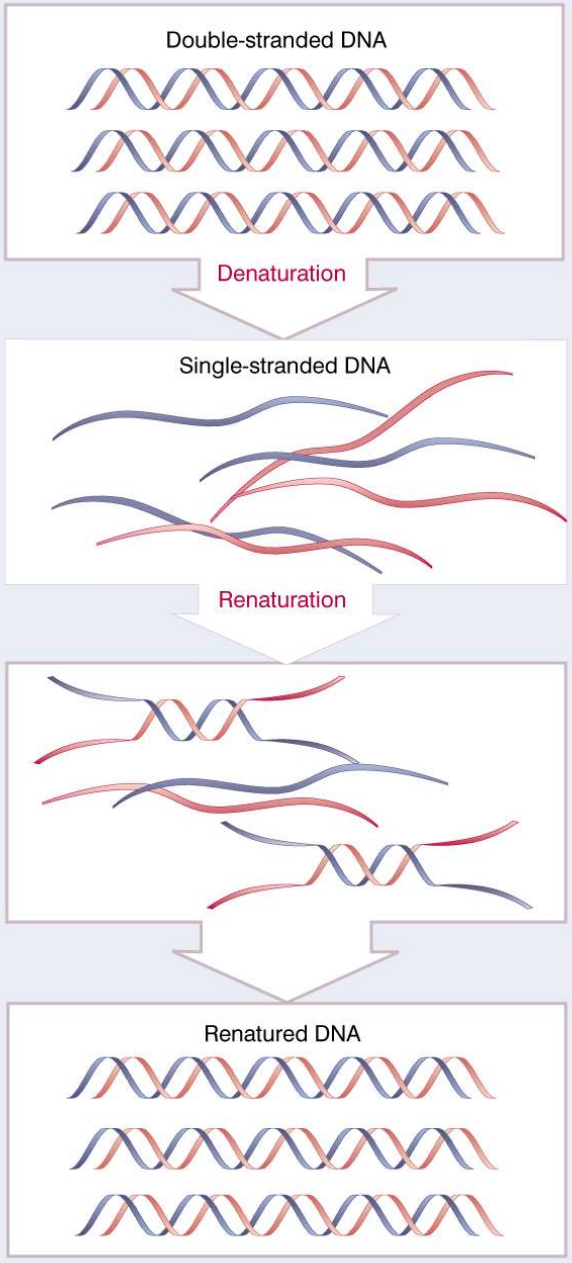
Desnaturação do DNA

Desnaturação
Temperatura > 85-90ºC

Soluções com valores de pH extremos
Soluções concentradas de formamida e ureia
Soluções com concentrações iónicas baixas
– baixam Tm

Renaturação
redução lenta da temperatura de 90 -> 65 ºC

Figure 1.13 Denatured single strands of DNA can renature to give the duplex form.



dsDNA –
DNA em cadeia dupla
forma NATIVA

ssDNA –
DNA em cadeia simples
forma desnaturada

A desnaturação de DNA pode ser monitorizada por espectrofotometria – 260 nm

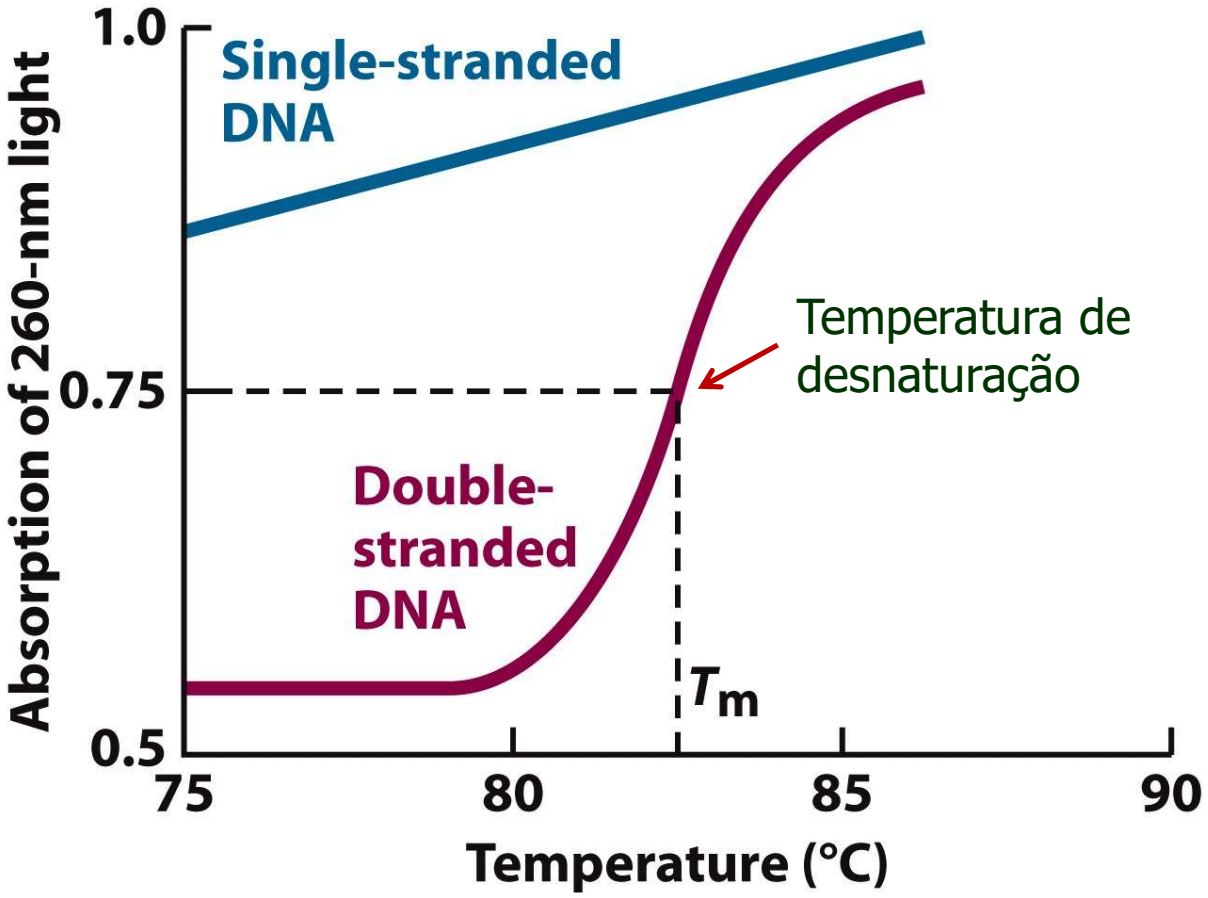
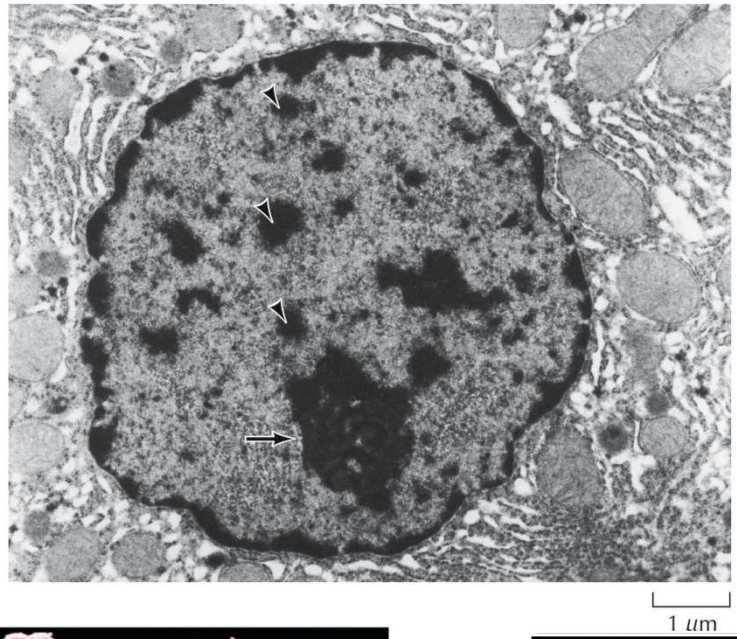
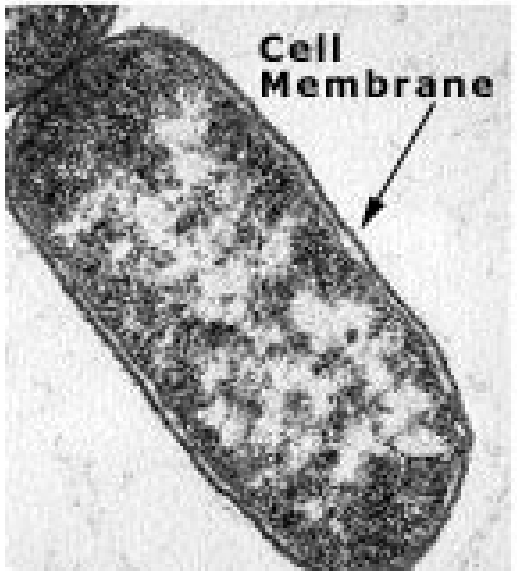


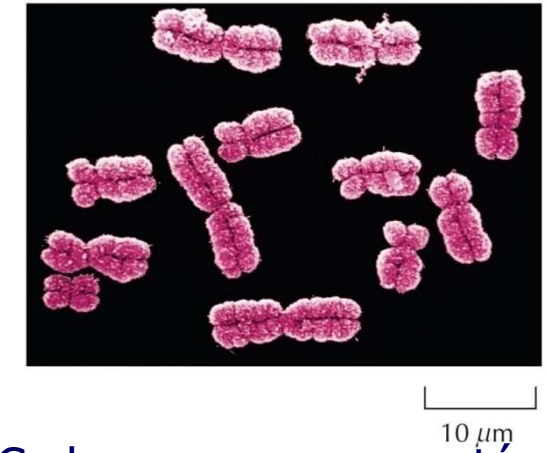
Figure 4-7a
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DNA no núcleo em interfase e durante a divisão celular

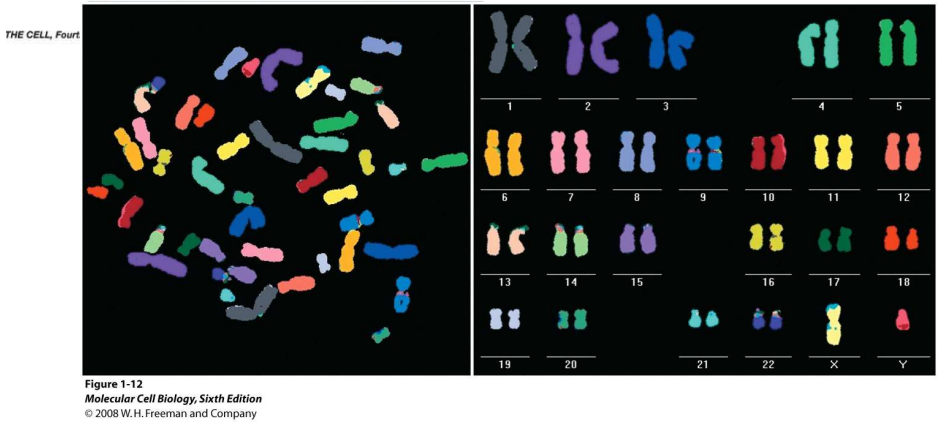




E. coli

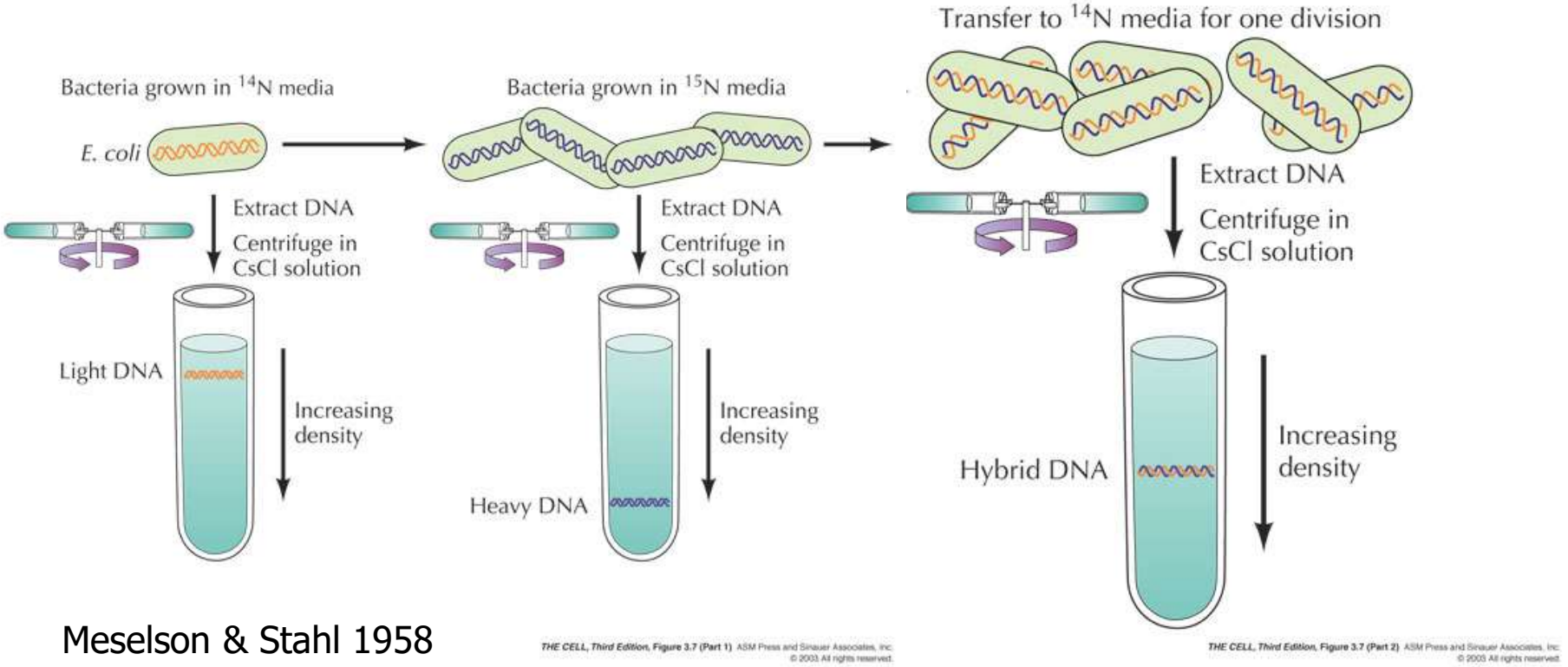


Cada cromossoma contém uma molécula de DNA linear



Cariótipo- Diagrama organizado dos cromossomas metafásicos de uma espécie

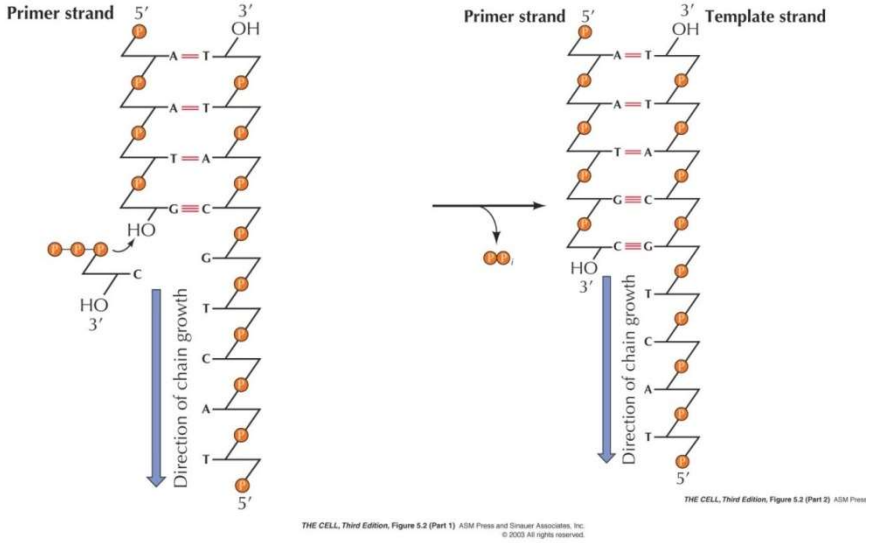
Replicação do DNA – Evidência experimental

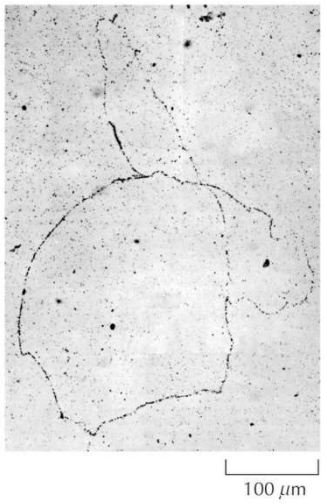


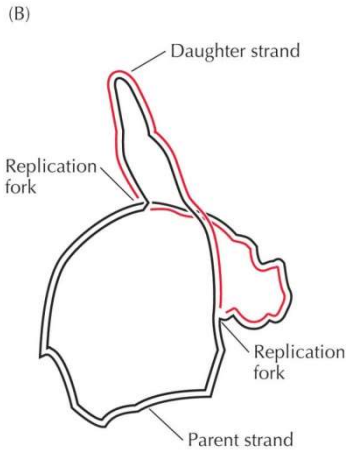
Meselson & Stahl 1958

A replicação do DNA é semi-conservativa

Reacção catalizada pela DNA Polimerase







O crescimento da forquilha de replicação é bi-direcional



As DNA polimerases sintetizam DNA sempre de 5´para 3´

As DNA polimerases são incapazes de sintetizar DNA *de novo*. Acrescentam nucleotídeos a uma região de cadeia dupla

A forquilha de replicação é assimétrica

Síntese das cadeias condutora e lenta do DNA

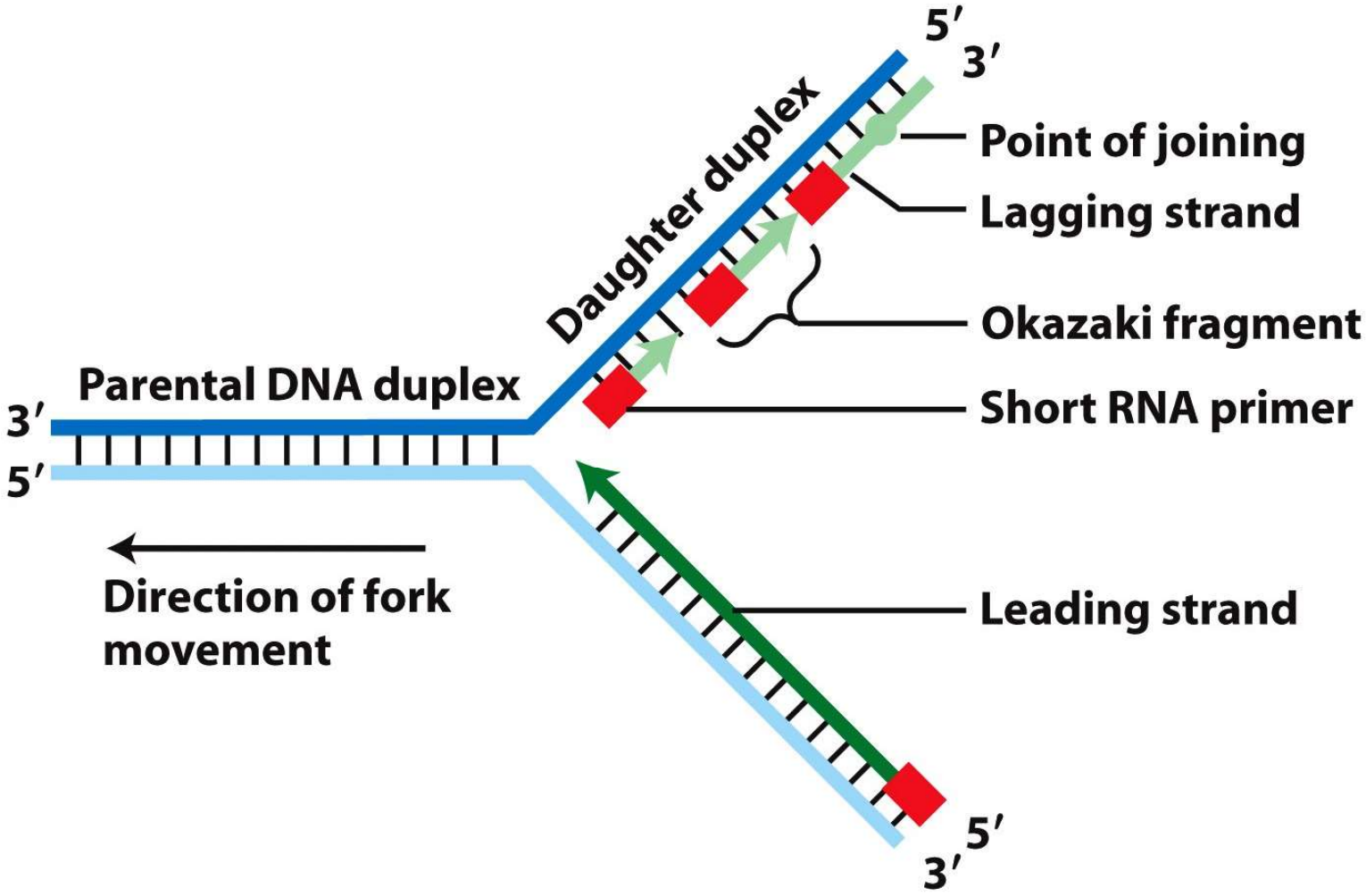
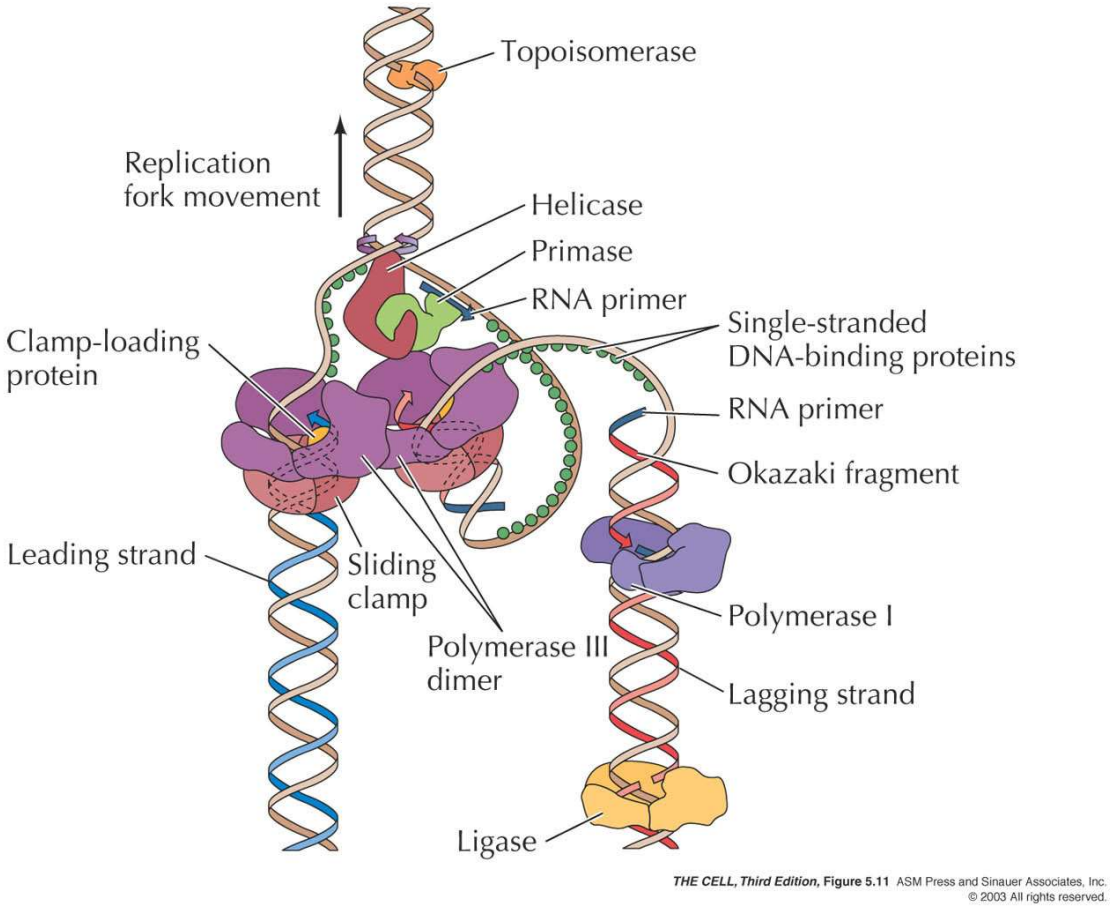
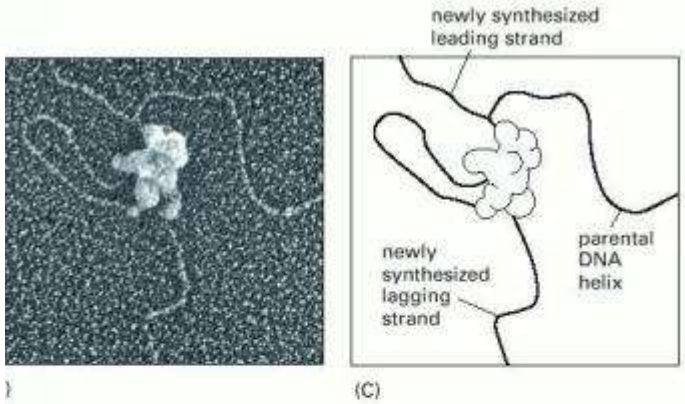


Figure 4-30
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Modelo da forquilha de replicação na *E. coli*

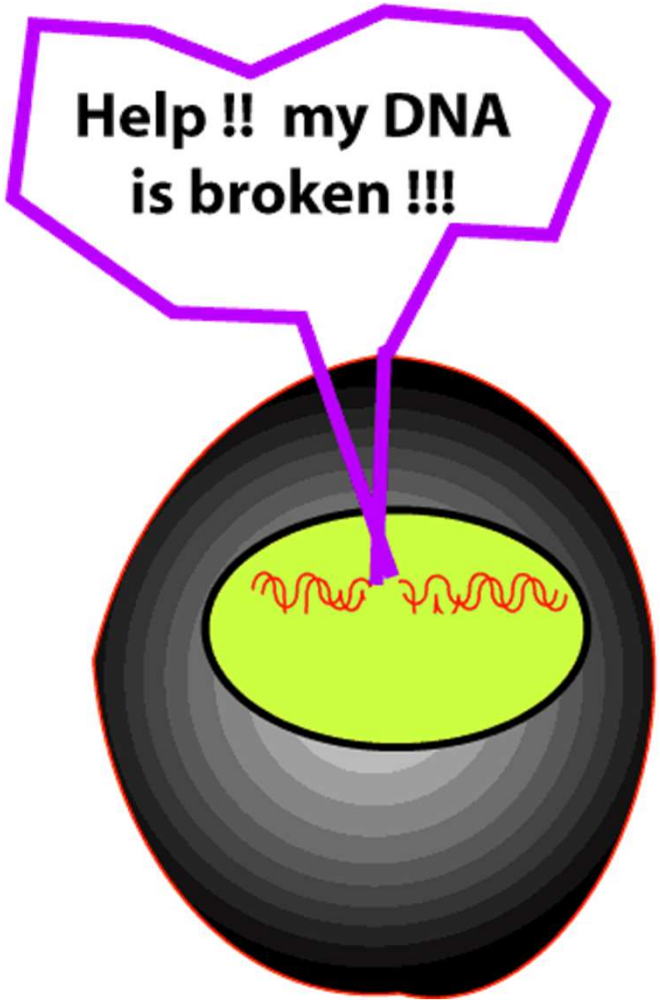


THE CELL, Third Edition, Figure 5.11 ASM Press and Sinauer Associates, Inc.
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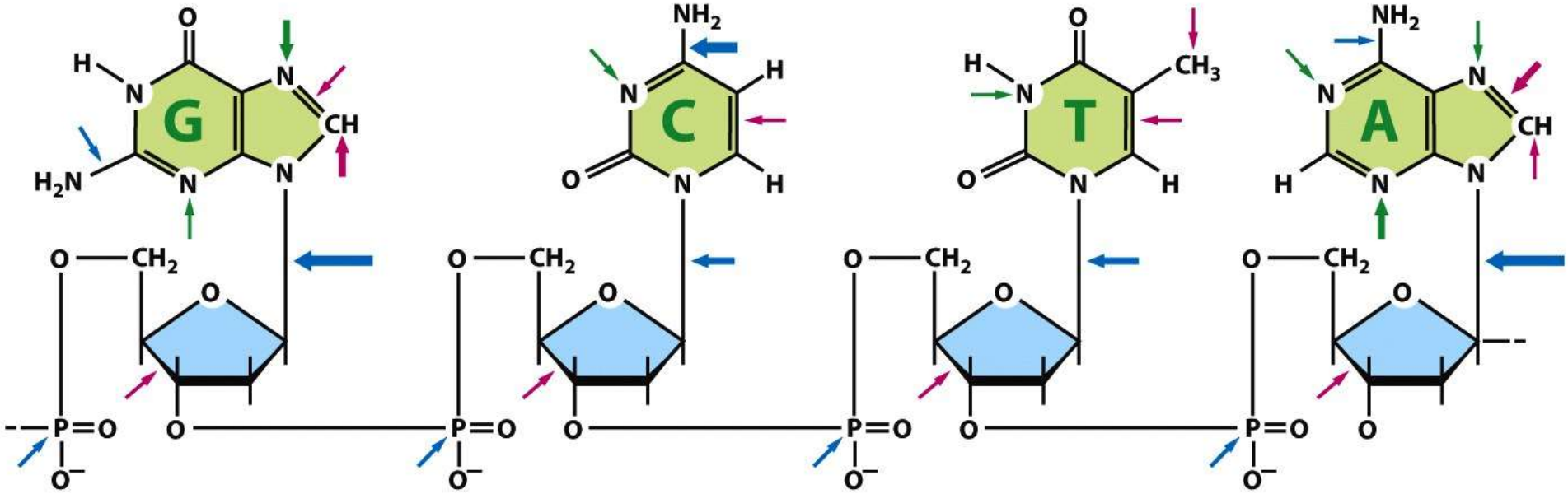


https://www.youtube.com/watch?v=vNXFk_d6y80

Cada célula humana sofre em média por dia 10^4 a 10^6 eventos conducentes a dano do DNA Lodish 6th ed p145



Modificações espontâneas no DNA que requerem reparação

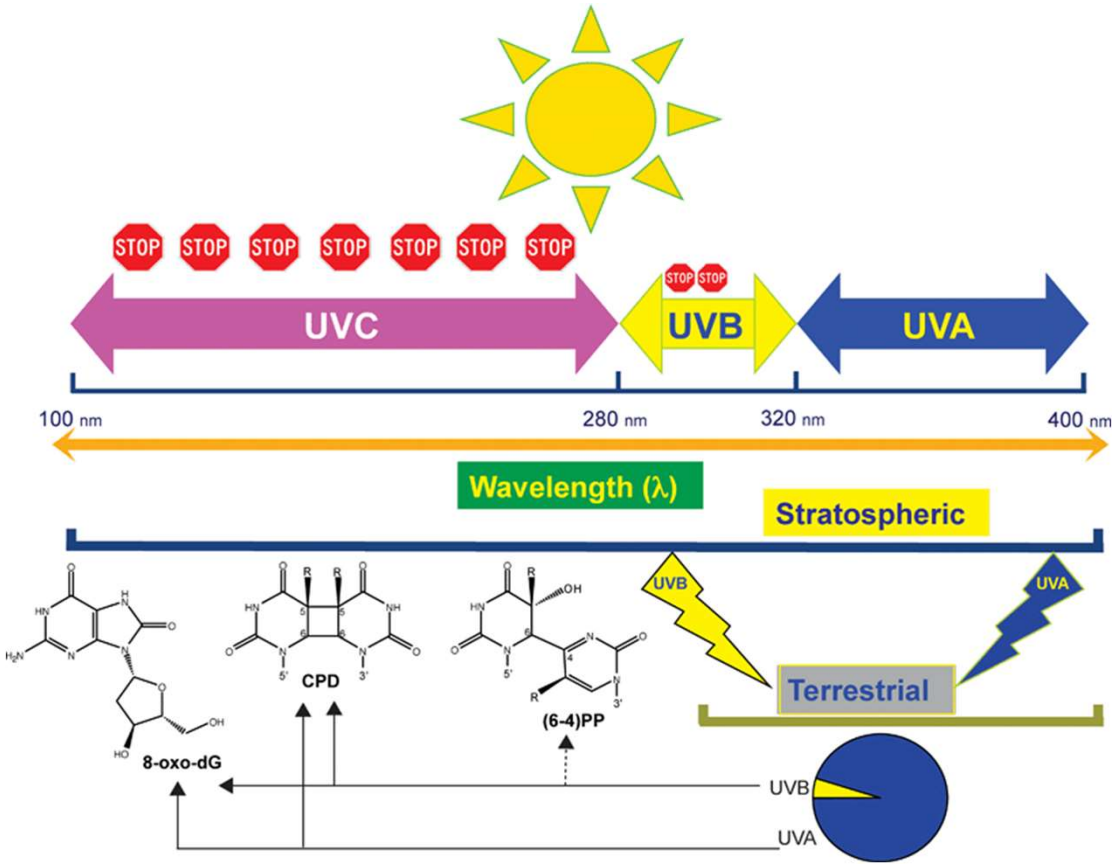


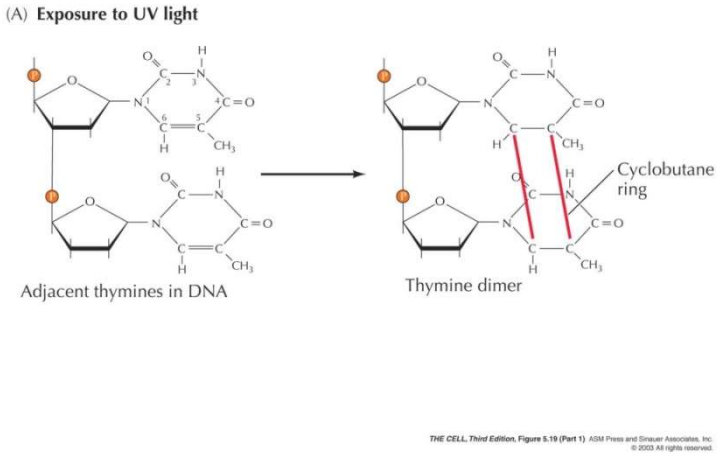
reacção de hidrólise

dano oxidativo

metilação

Fatores extrínsecos
Mutações no DNA induzidas por radiações

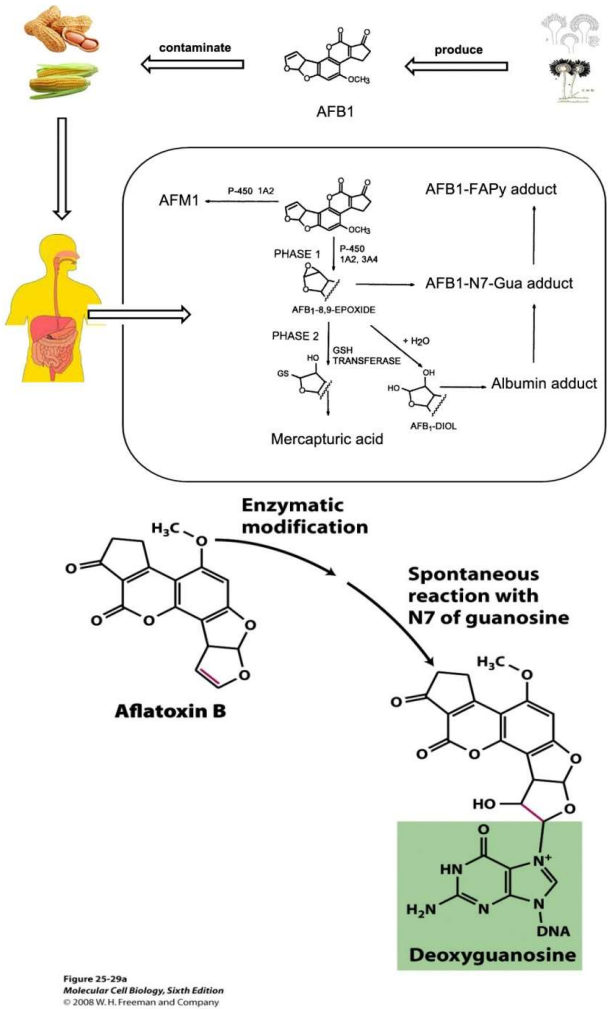


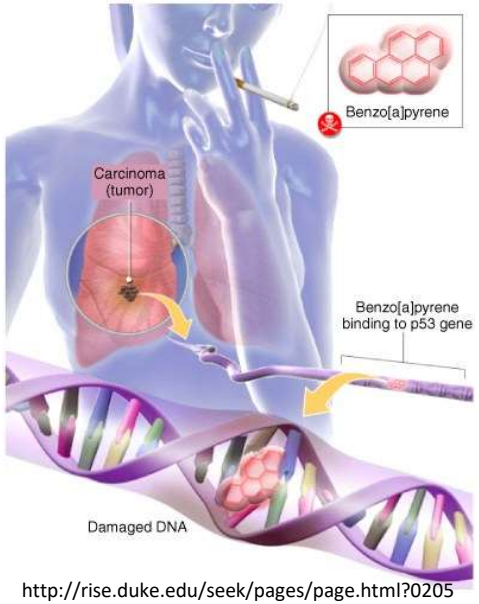


http://pubs.rsc.org/en/content/articlehtml/2012/pp/c1pp05144j

Fatores extrínsecos

Mutações no DNA induzidas por compostos químicos





http://rise.duke.edu/seek/pages/page.html?0205

Table 1 Representative DNA adducts of some cigarette smoke carcinogens and related compounds: occurence and likely consequent mutations

Carcinogen[a]	DNA Base	Adduct[b]	Detected in smokers' lung? (type of evidence)[c]	Likely mutations	Reference
BaP	G		Yes (2)	GC→TA	(Seo, et al. 2000; Yoon et al., 2001)
5-MeC	G		No	GC→TA GC→CG	(Misra et al., 1998)
NDMA NNK	G		Yes (1) Yes (1)	d GC→AT	(Zhao et al., 1999) (Wilson et al., 1988; Loechler et al., 1984)
NNK, NNN	G		Yes (2)	GC→TA, GC→AT	(Ronai et al., 1993; Folies et al., 1991)
NPYR	G		No		(Wang et al., 2001b)
Ethylene oxide	G		Yes (1)	d	((Zhao et al., 2000)
1,3-butadiene	G		No	d	(Kloc et al., 1999; Tretyakova et al., 1997)
	G		No	d	(Kloc et al., 1999; Tretyakova et al., 1997)
	A		No		(Tretyakova et al., 1997)
	A		No		(Tretyakova et al., 1997)
	G		No	d	(Kolmisto et al., 1999; Kloc et al., 1999)
	A		No		(Zhao et al., 2000)
	G		No	GC→TA>GC→AT>GC→CG	(Carmical et al., 2000)
Acetaldehyde	G		No		
	G		No	GC→AT, AT→TA	(Nicoll and Hou, 2001; Wang et al., 2000a)
Crotonaldehyde	G		No		(Chung et al., 1999)
	G		No		(Wang et al., 2001a)
4-aminobiphenyl	G		Yes (1,2)	GC→TA, GC→CG	(Culp et al., 1997; Lin et al., 1994; Maher Jr. et al., 1994)
Vinyl chloride	G		No	d	(Swenberg et al., 1999)
	G		No	GC→AT	(Nair et al., 1999)
Vinyl chloride Ethyl carbamate	A		No	AT→GC AT→TA AT→CG	(Nair et al., 1999; Guengerich and Kim, 1991)
2-nitropropane	G		No		(Sodum and Fiala, 1998)
Oxidants	G		Yes (3)	GC→TA GC→AT	(Asami et al., 1997; Moriya, 1993)
Nitric oxide	G		No		(Burney et al., 1999)
Peroxynitrite	8-oxo-G		No	GC→TA	(Henderson et al., 2002)
			No	GC→TA	
			No	GC→TA	

[a] Abbreviations: BaP, benz[a]pyrene; 5-MeC, 5-methylchrysene; NDMA, *N*-nitrosodimethylamine; NNK, 4-(methylnitrosamino)-1-(3-pyridyl)-1-butanone; NNN, *N*-nitrosonornicotine; NPYR, *N*-nitrosopyrrolodine

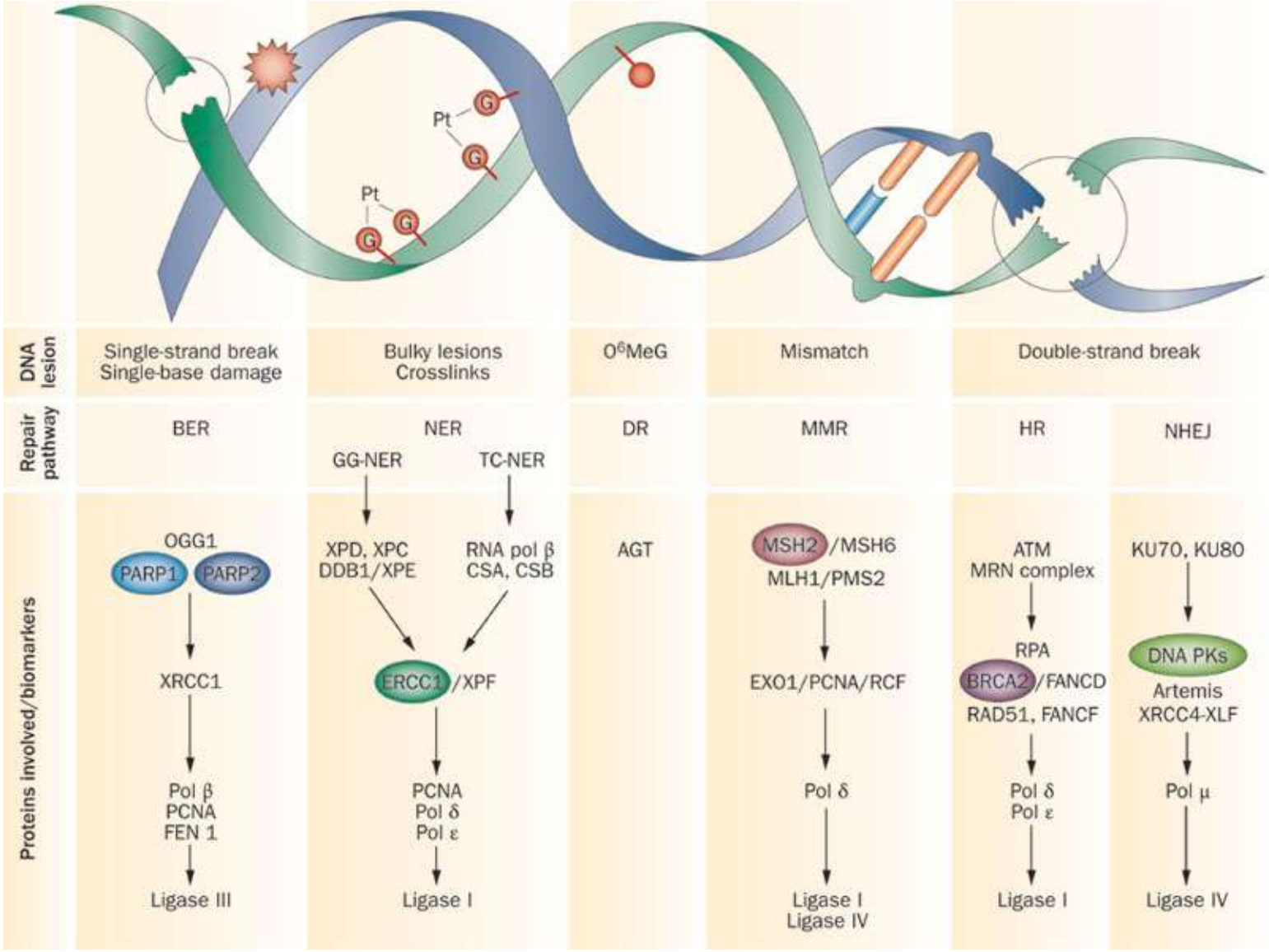
[b] Adduct structures show the position of attachment to the base (e.g. N²-, O⁶-, or 7- of G) and the organic moiety derived from the carcinogen.

[c] Type of evidence
3. detection of nucleoside or base by specific method (e.g. MS, HPLC fluorescence, or HPLC-electrochemical detection)
2. detection of released adducting moiety by specific method
1. detection of nucleoside or base by relatively non-specific method (e.g. 32P-post-labelling or immunoassay)

[d] Likely to give GC→TA transversions via depurination to an apurinic site

Toxina produzida pelos fungos *Aspergillus parasiticus* e *Aspergillus flavus* que existem no amendoim. Aumenta risco de cancro no fígado

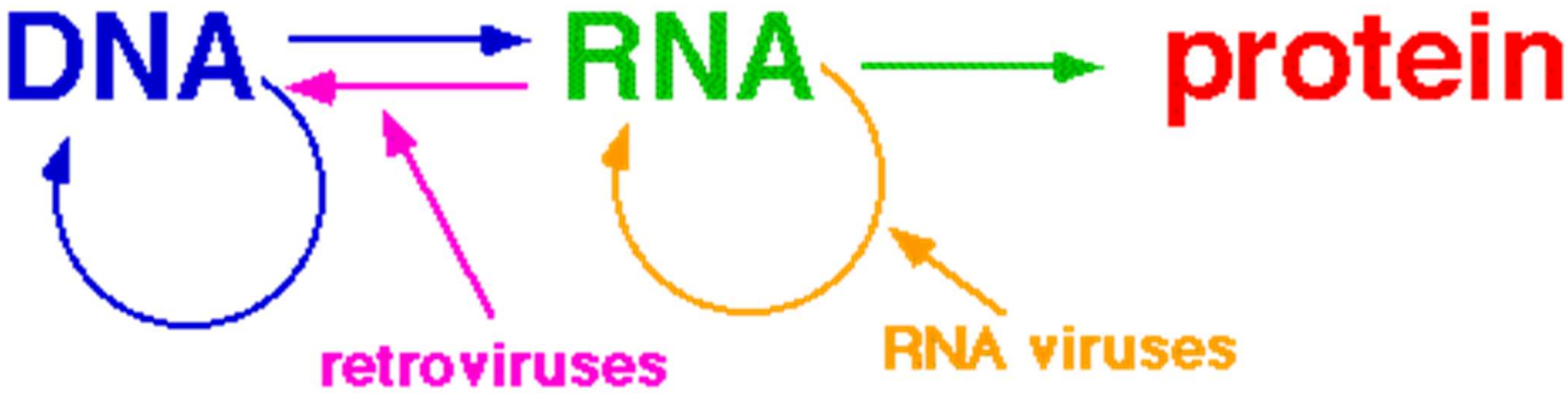
http://www.nature.com/onc/journal/v21/n48/full/1205803a.html



http://www.nature.com/nrclinonc/journal/v9/n3/fig_tab/nrclinonc.2012.3_F1.html

Dogma central da Biologia Molecular

Francis Crick - 1958 *The central dogma of molecular biology deals with the detailed residue-by-residue transfer of sequential information. It states that such information cannot be transferred back from protein to either protein or nucleic acid.*



Gene – sequência de DNA que contém a informação necessária para a síntese de uma cadeia polipeptídica ou um RNA funcional

Transcrição do DNA

A polimerização do RNA ocorre sempre de 5´para 3´

As RNA polimerases podem iniciar a polimerização de ribonucleotídeos sem primers

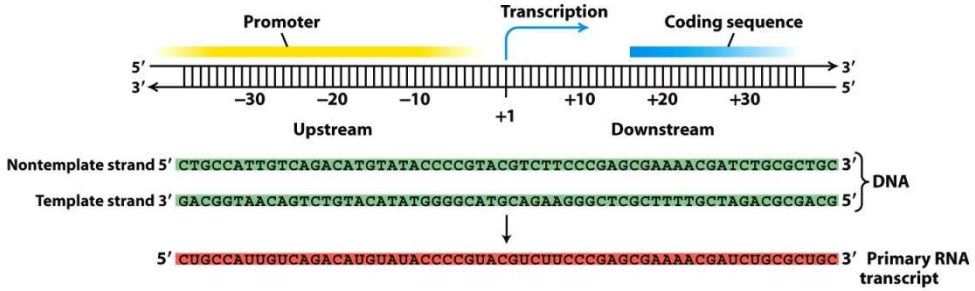


Figure 4-10b
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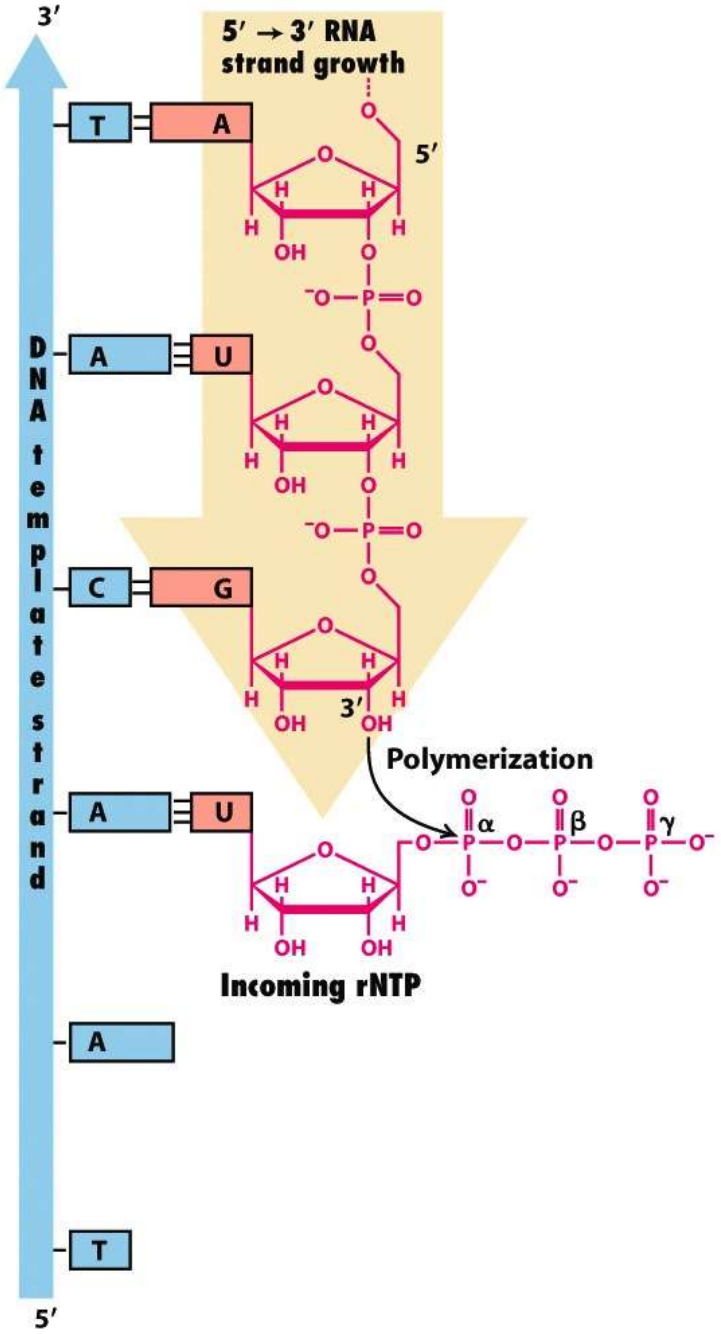
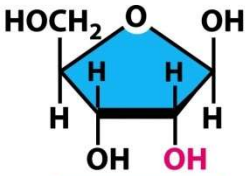


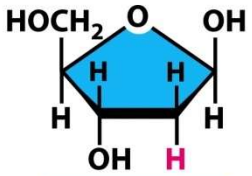
Figure 4-10a
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RNA



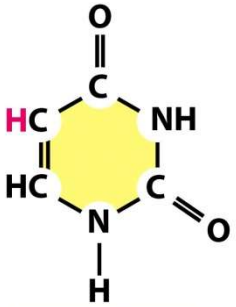
ribose

used in ribonucleic acid (RNA)



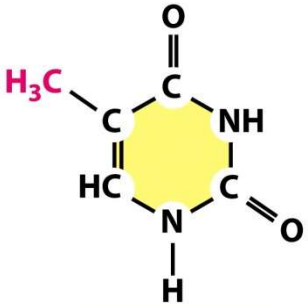
deoxyribose

used in deoxyribonucleic acid (DNA)



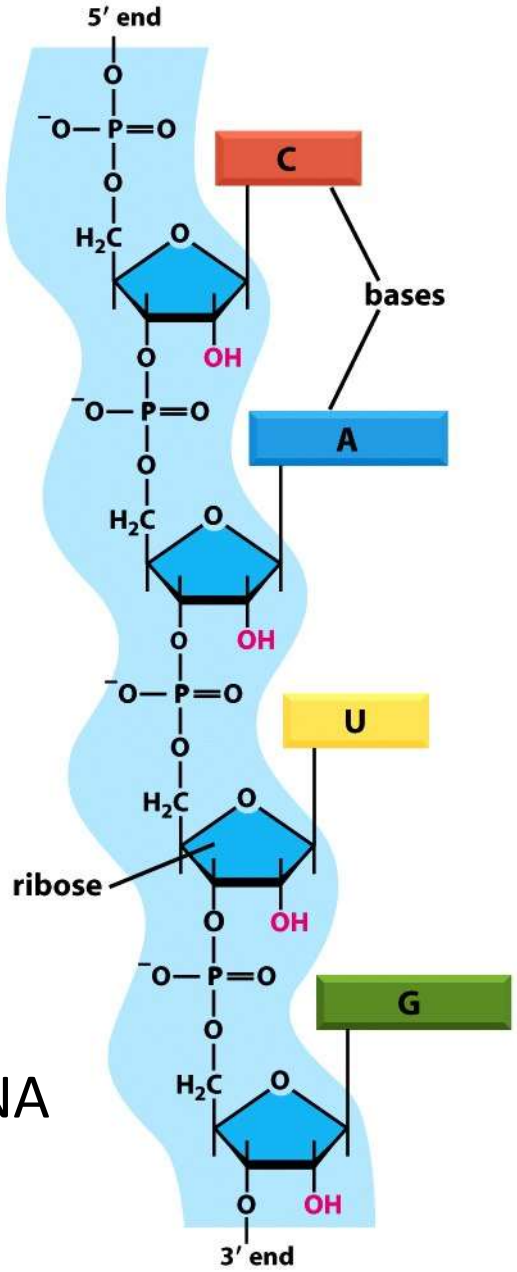
uracil

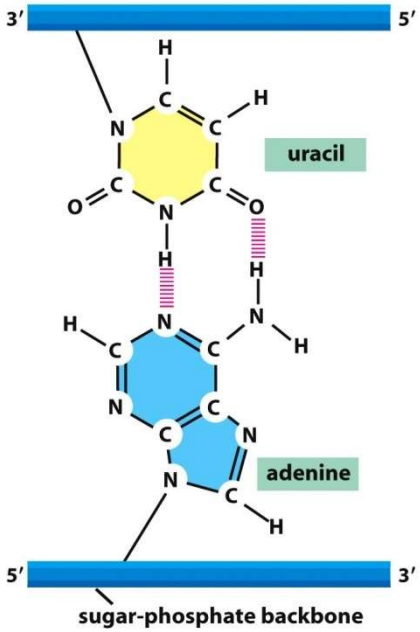
used in RNA



thymine

used in DNA

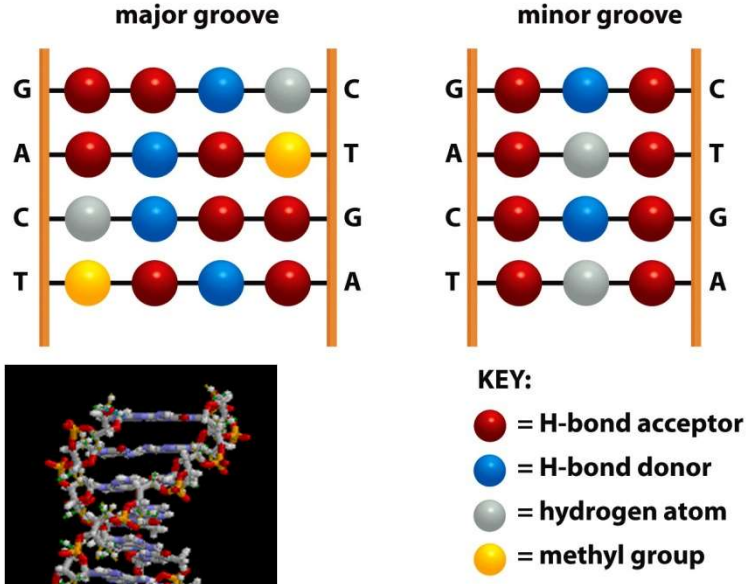


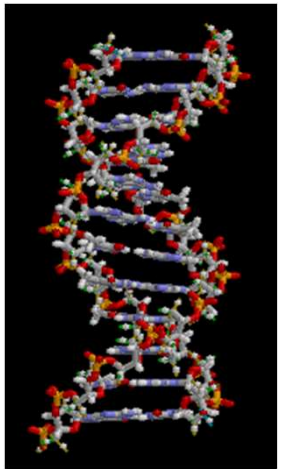


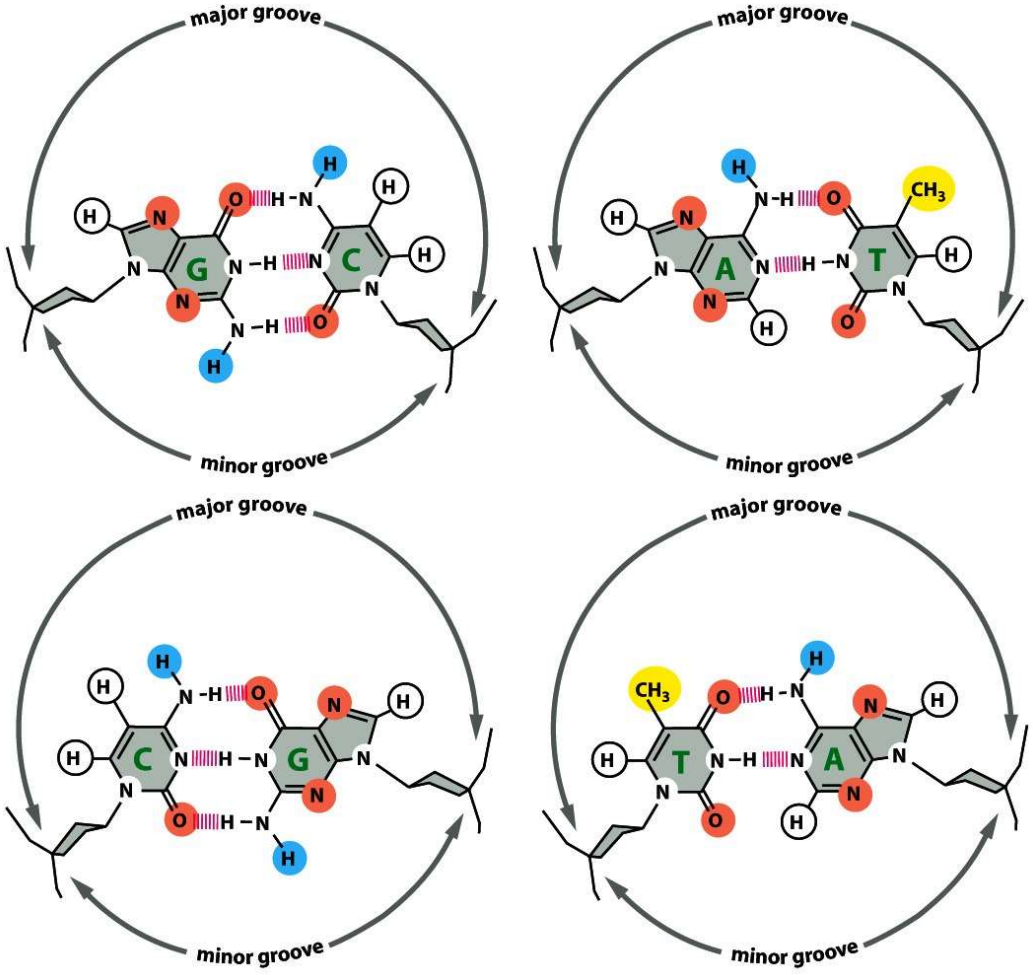
As cadeias simples de DNA e RNA podem emparelhar entre si por complementaridade de bases

A sequência de bases pode ser identificada por proteínas a partir da periferia da molécula de DNA

As proteínas reguladoras estabelecem ligações com as bases expostas, em especial, no sulco maior da cadeia de DNA







Interações comuns entre o DNA e as proteínas

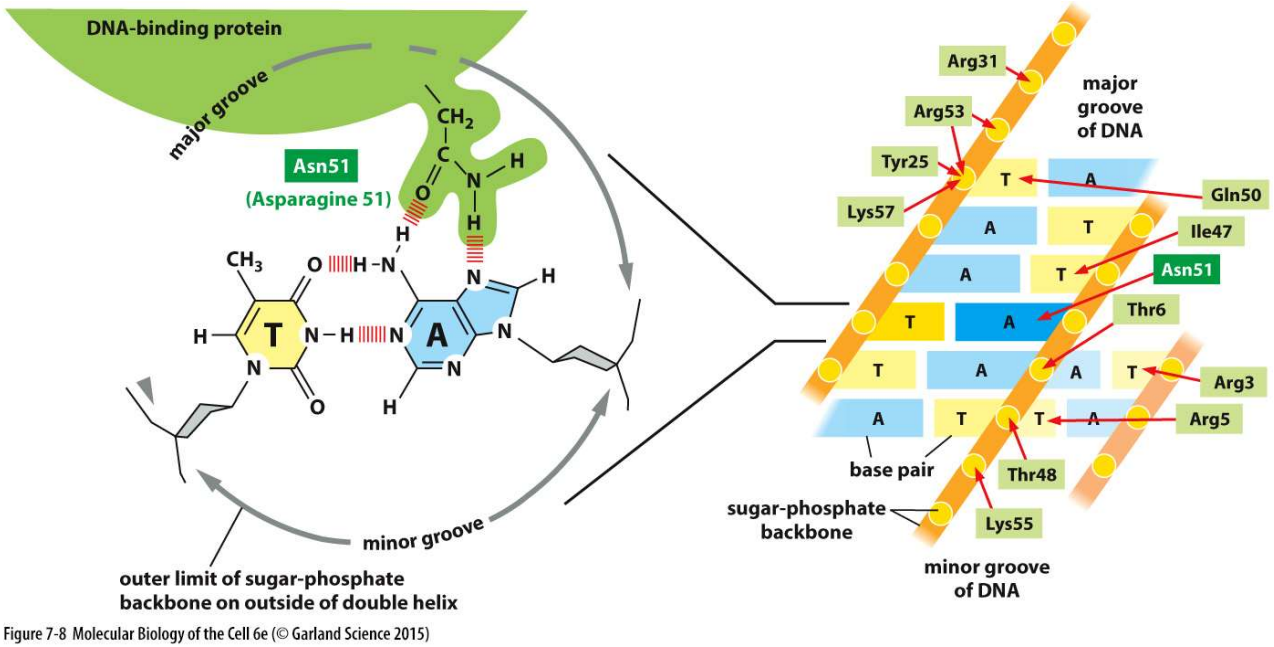
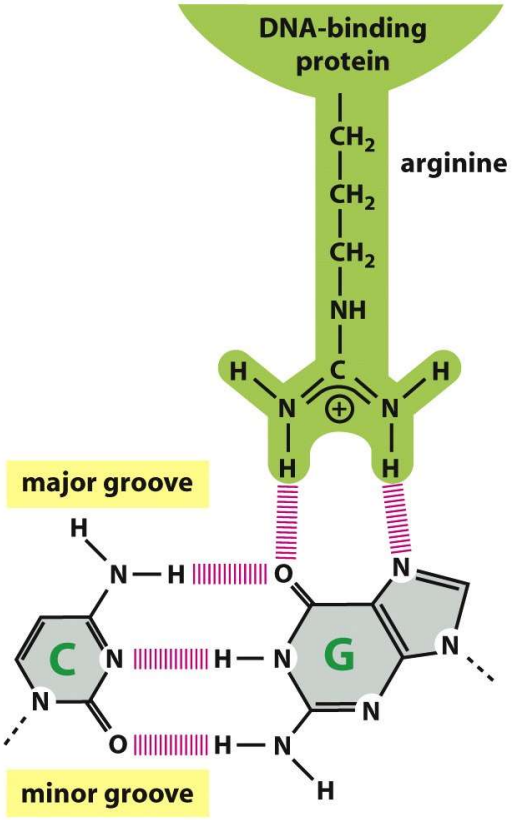


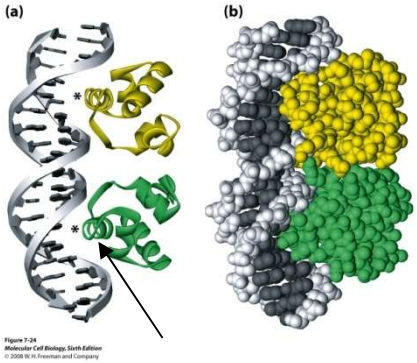
Figure 7-8 Molecular Biology of the Cell 6e (© Garland Science 2015)



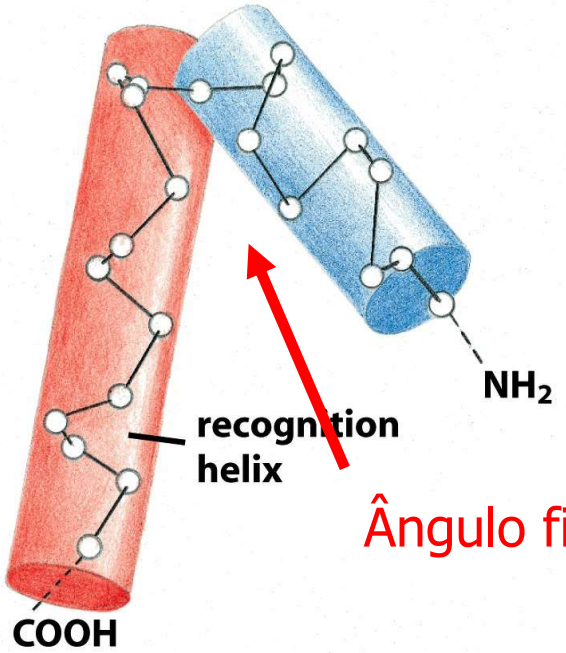
Em cada local de reconhecimento de uma sequência de nucleotídeos por uma proteína estabelecem-se cerca de 20 contactos na interface DNA-proteína

Domínios de ligação ao DNA presentes nas *DNA-binding proteins*

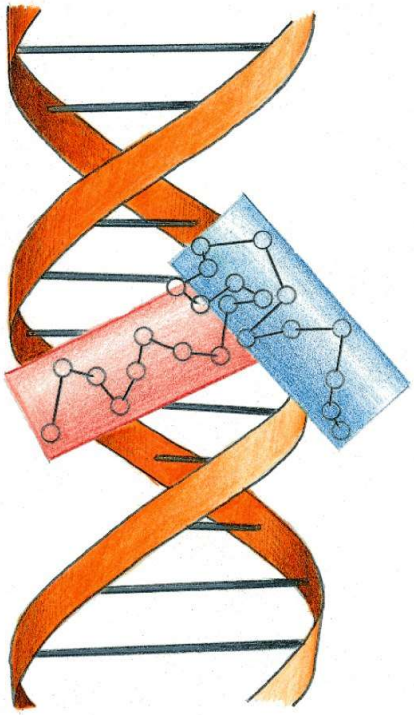
Helix-turn-helix



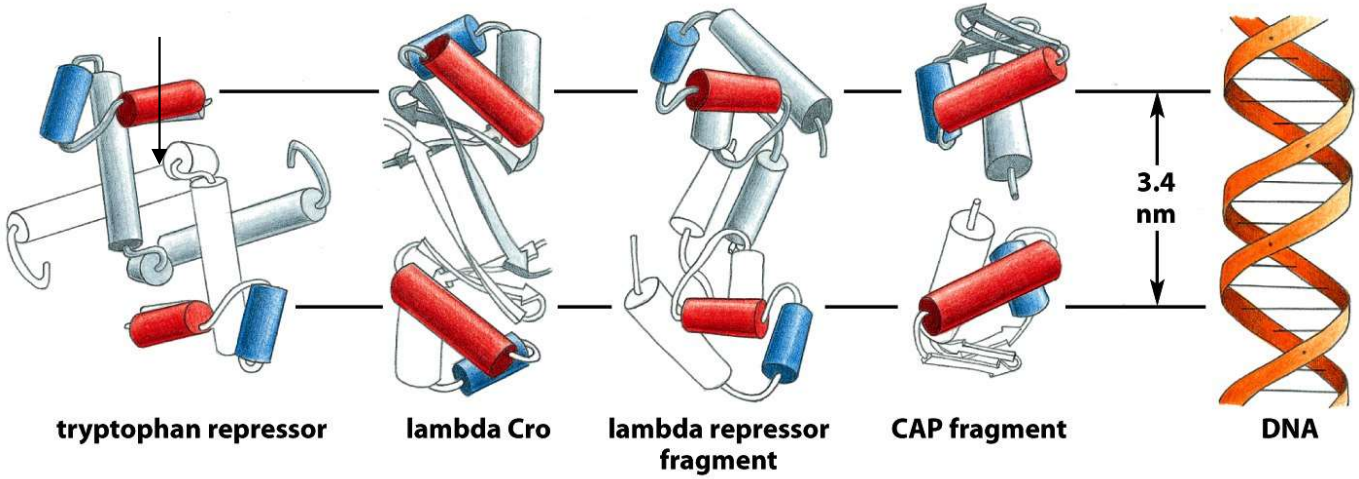
Hélice de reconhecimento



Ângulo fixo



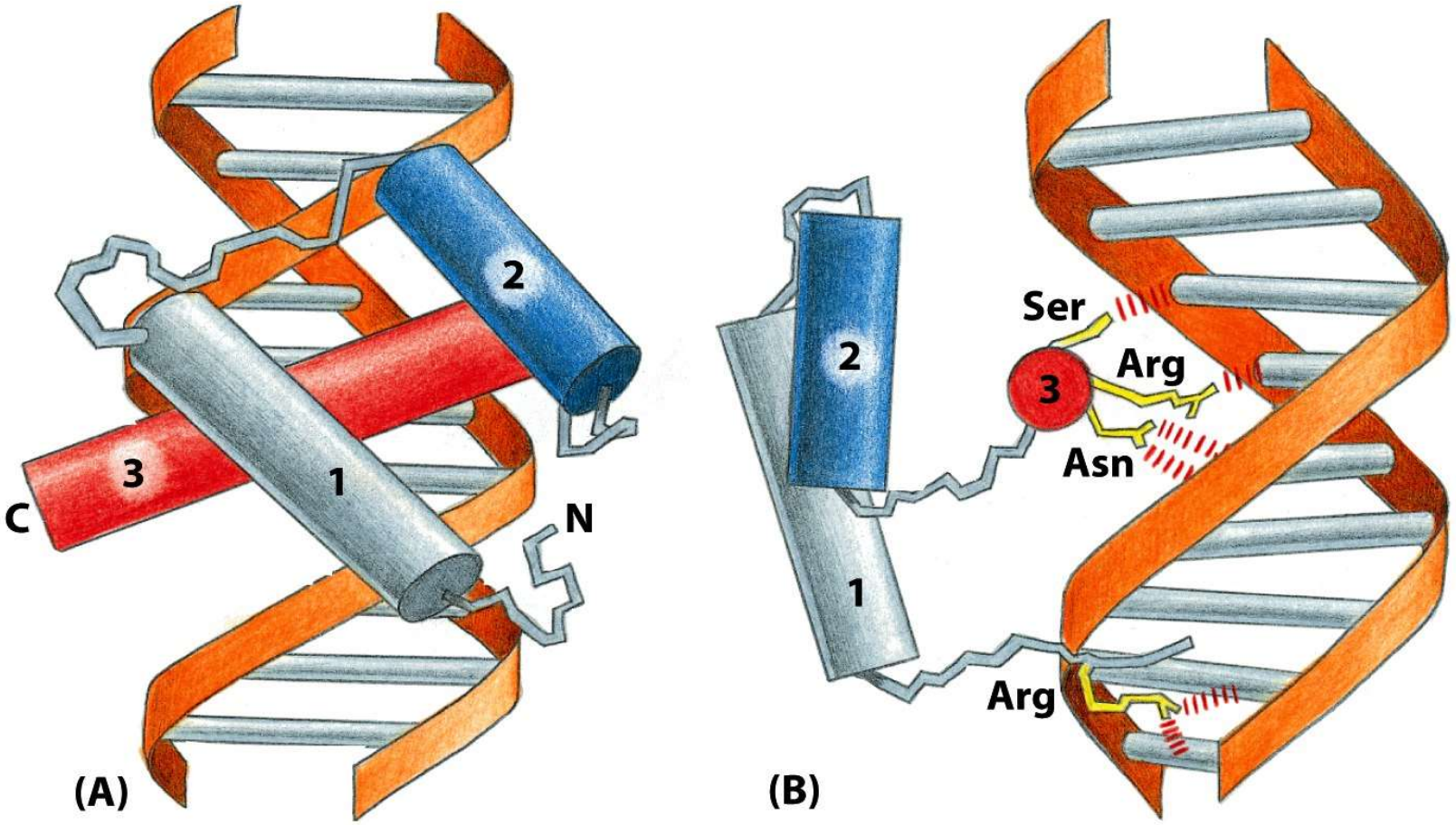
Hélice de reconhecimento (A) (B)



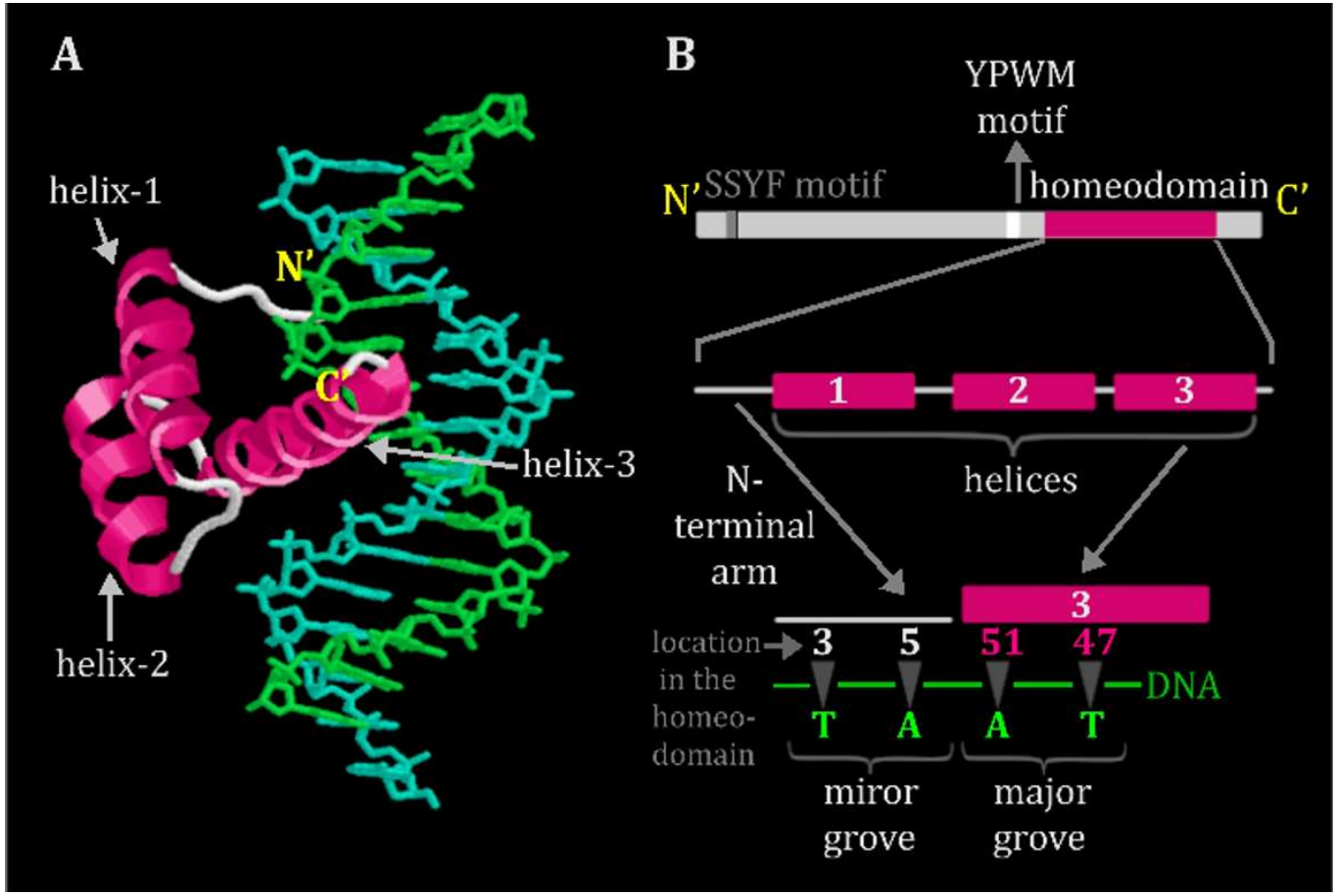
Dímeros simétricos separados de uma distância de uma volta – turn da hélice de DNA

Homeodomain presente nas células eucarióticas -

um caso particular de helix-turn-helix observada nos repressores bacterianos



Proteínas Hox



http://bioinformatics.uni-konstanz.de/HueberHox/Atlas/Hox_proteins.php

http://www.youtube.com/watch?v=JgVjCFdoEQc

Domínios de ligação ao DNA presentes nas *DNA-binding proteins*

Zinc-Finger é o mais abundante motivo de ligação ao DNA nas células animais

Gl1
C2H2 zinc finger

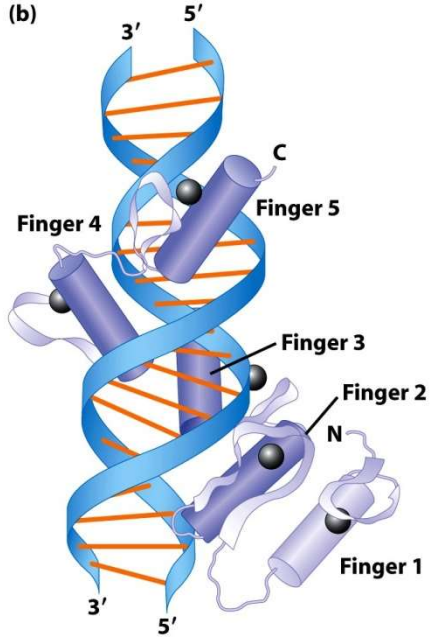
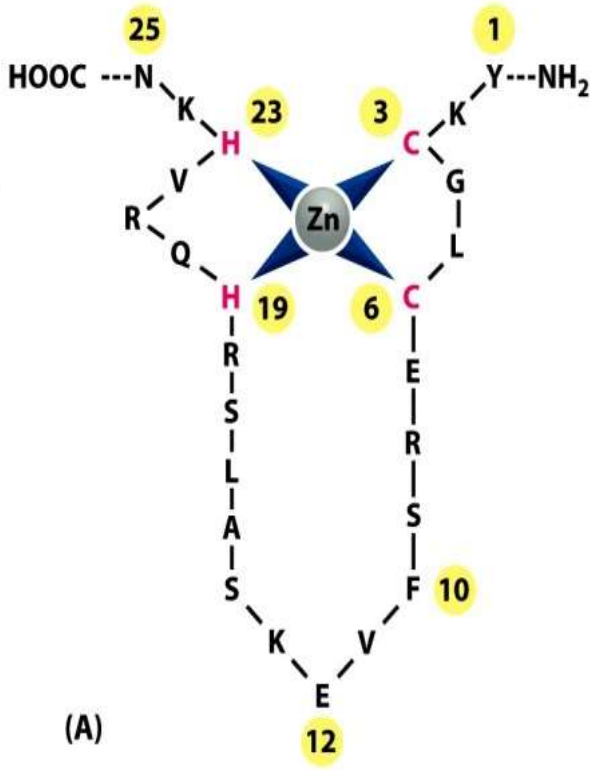
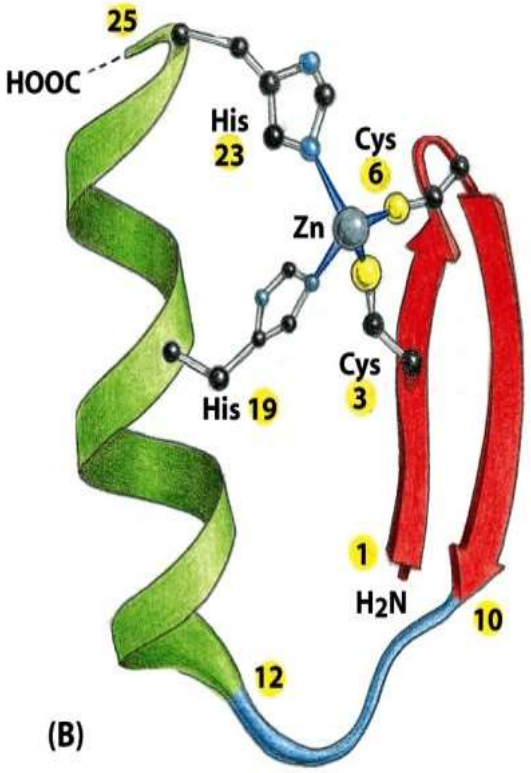


Figure 7-25bc
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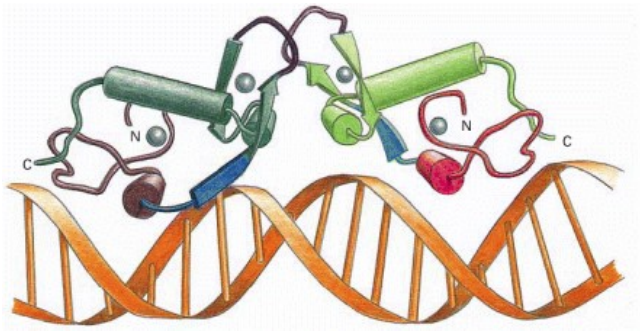


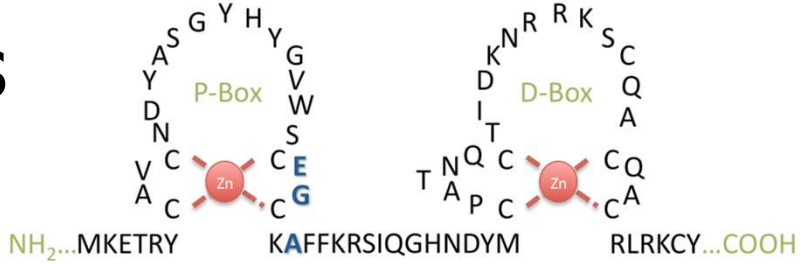
http://www.youtube.com/watch?v=wPtK9NxbGPw&feature=relmfu

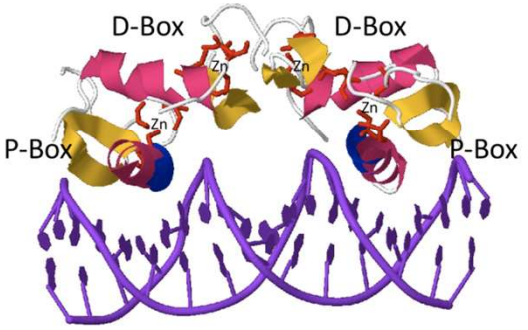
Domínios de ligação ao DNA presentes nas *DNA-binding proteins*

Zinc-Finger do tipo C4

Pertencem à super-família dos recetores nucleares





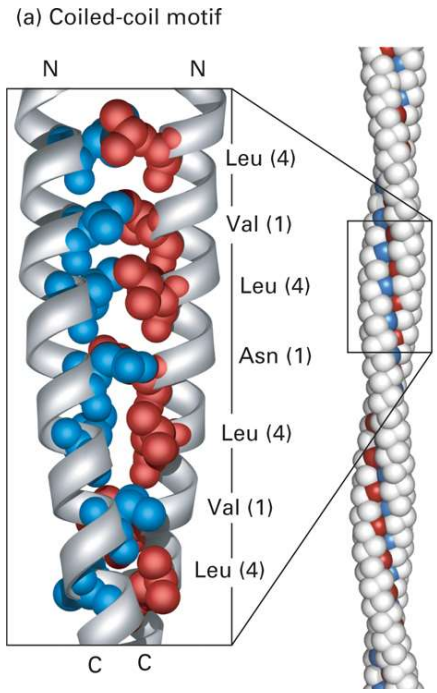


RECEPTORS	P-BOX	HALF-SITE	RESPONSE ELEMENT
ER	c**EG**ck**A**	AGGTCA	AGGTCAnnnTGACCT
GR, MR, PR, AR	c**GS**ck**V**	TGTTCT	AGGACAnnnTGTCCT
PPAR, RAR, VDR, PPAR	c**EG**ck**G**	AGGTCA	AGGTCAnAGGTCA

NRR
Nuclear Receptor Resource

Domínios de ligação ao DNA presentes nas *DNA-binding proteins*

Leucin-zipper



P. ex. o factor de transcrição GCN4 (General Control Nonderepressible) da levedura

Basic helix-loop-helix (bHLH)

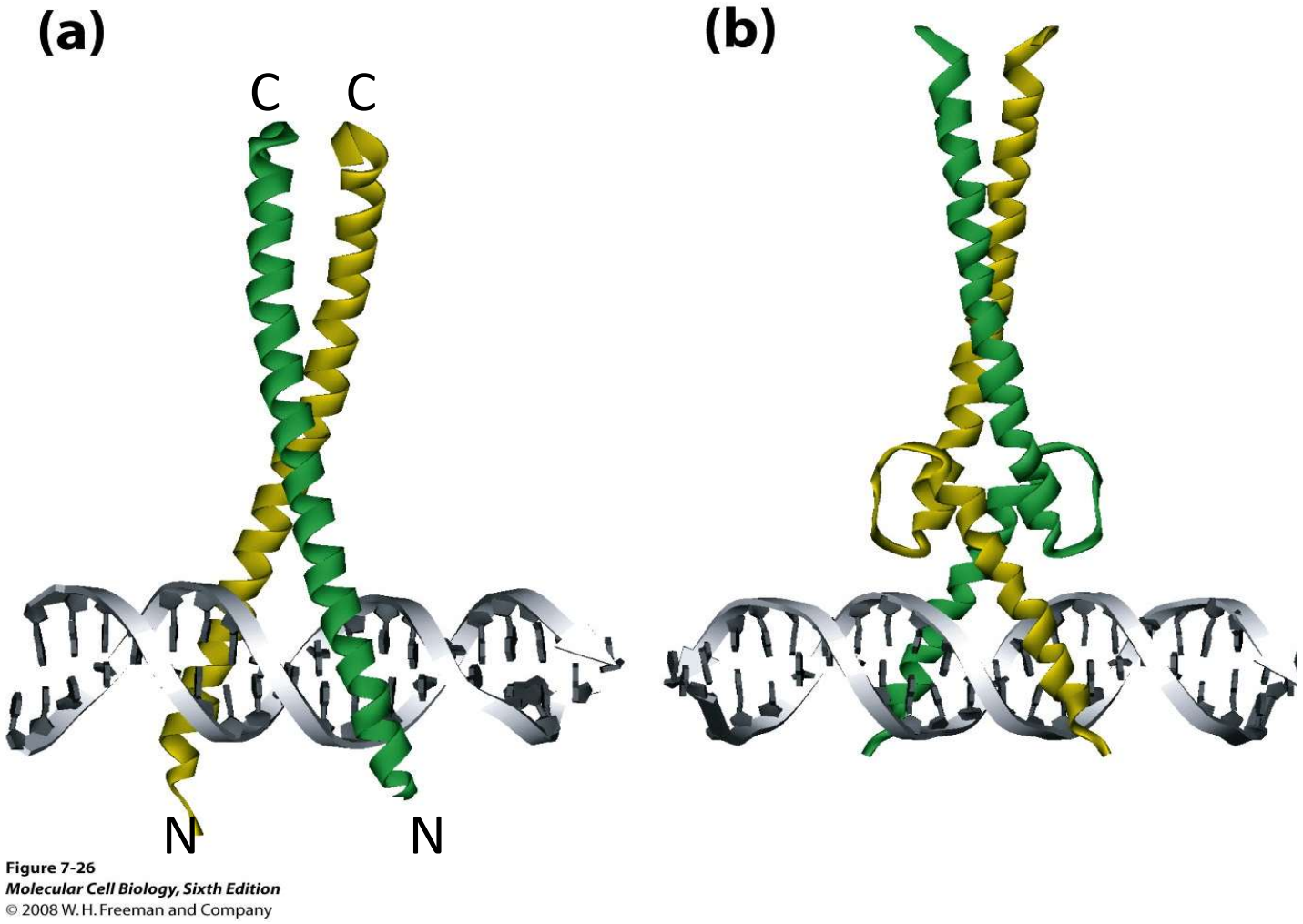
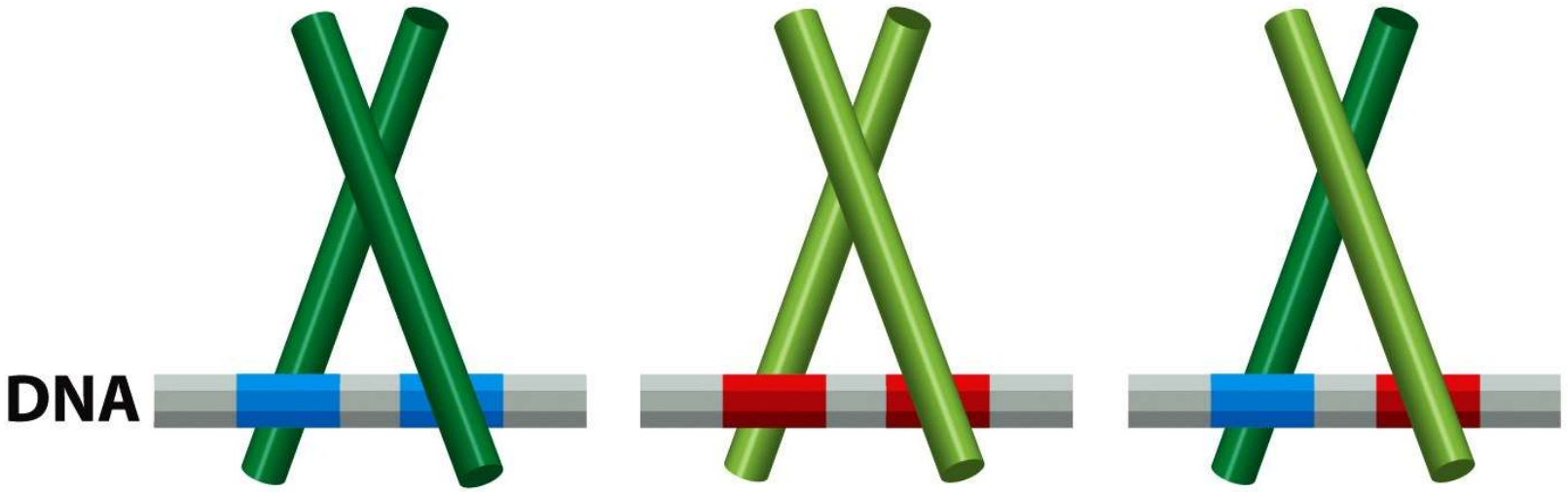


Figure 7-26
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http://www.youtube.com/watch?v=B5E1AXSPCFM

Domínios de ligação ao DNA presentes nas *DNA-binding proteins*

O Leucin-zipper e as hélice-ansa-hélice básicas podem formar diferentes heterodímeros com diferentes especificidades de ligação ao DNA

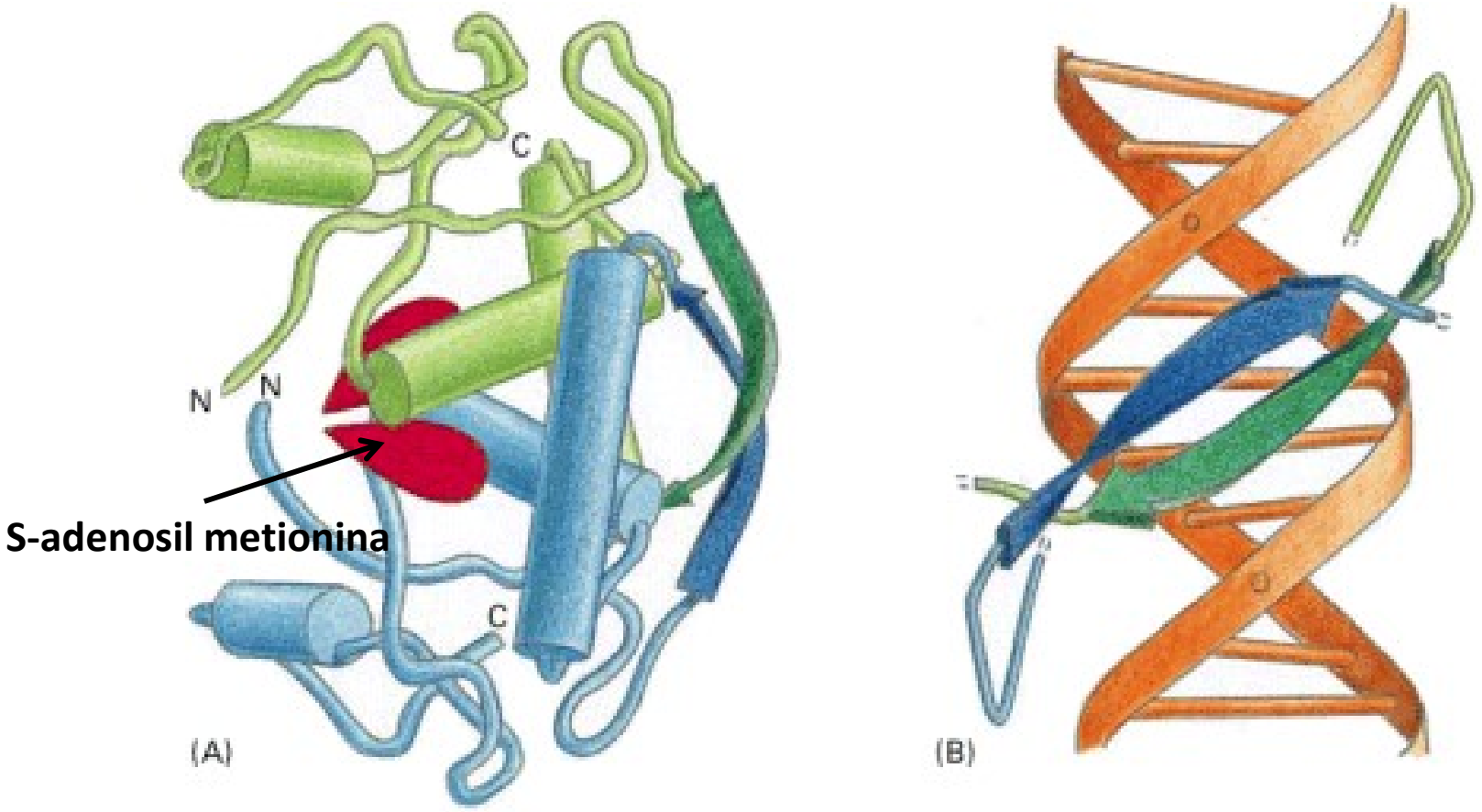


no entanto, o número de combinações em cada célula é limitado, dependendo sempre da sequência de aminoácidos de cada cadeia.

Domínios de ligação ao DNA presentes nas *DNA-binding proteins*

As β-sheets também conseguem reconhecer o DNA

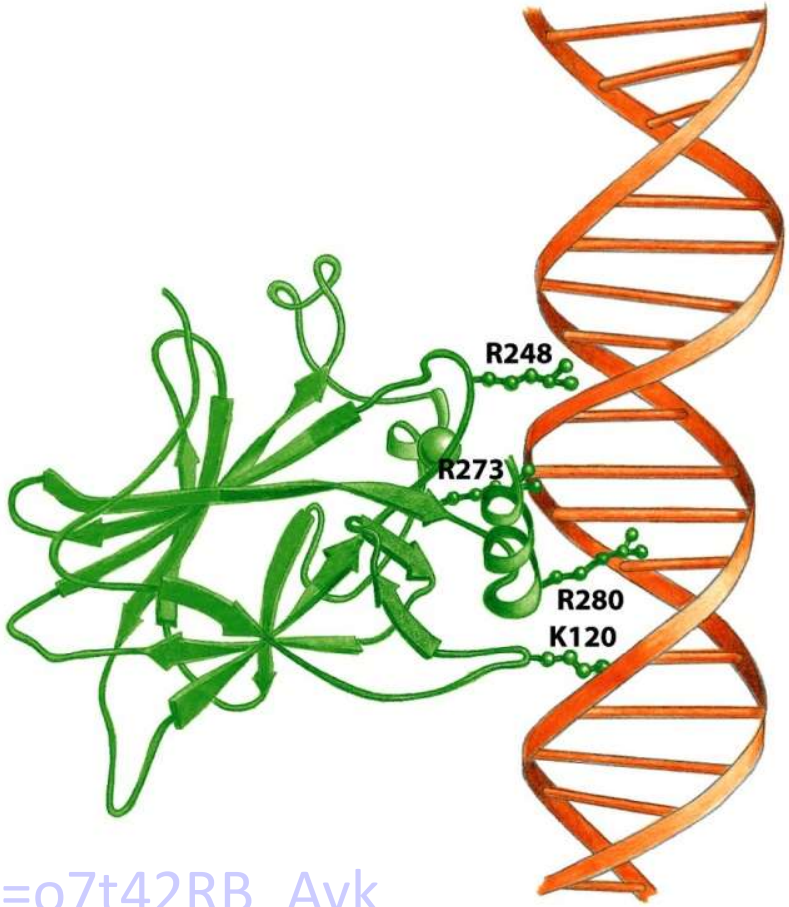
Proteína repressora *met* da bactéria



Domínios de ligação ao DNA presentes nas *DNA-binding proteins*

As sequências de bases nucleotídicas na molécula de DNA podem ser reconhecidas por ansas polipeptídicas

Proteína p53



https://www.youtube.com/watch?v=o7t42RB_Ayk

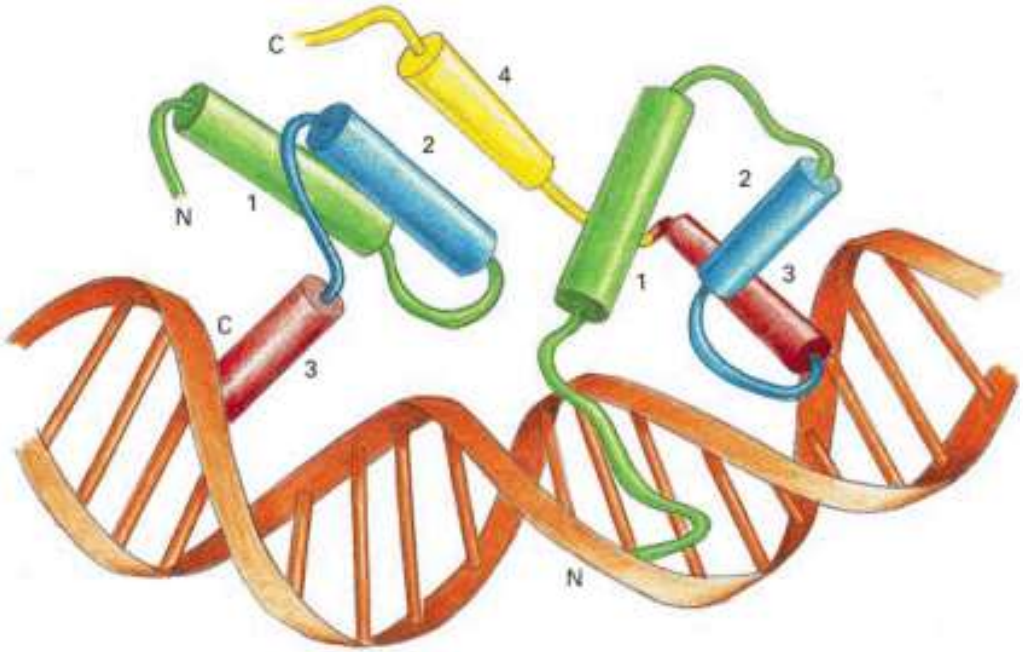
Mutações na arginina 248 são a mais frequente mutação da p53 conducente a cancro.

Domínios de ligação ao DNA presentes nas *DNA-binding proteins*

Também se podem formar heterodímeros

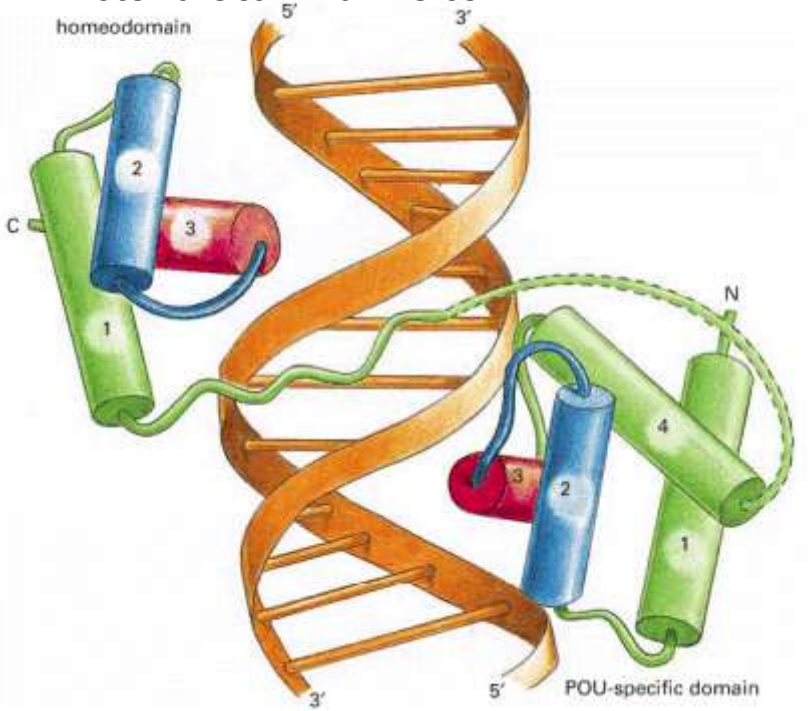
de homeodomains

Matα2 e Matα1 da levedura

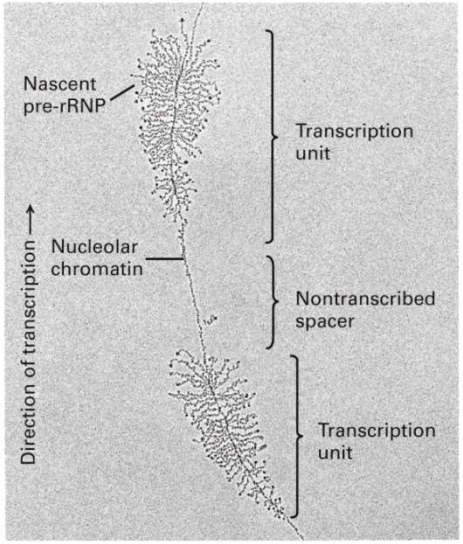


que até podem ser ligados por uma cadeia peptídica

Proteína Oct 1 mamíferos



A transcrição do DNA tem lugar no núcleo celular



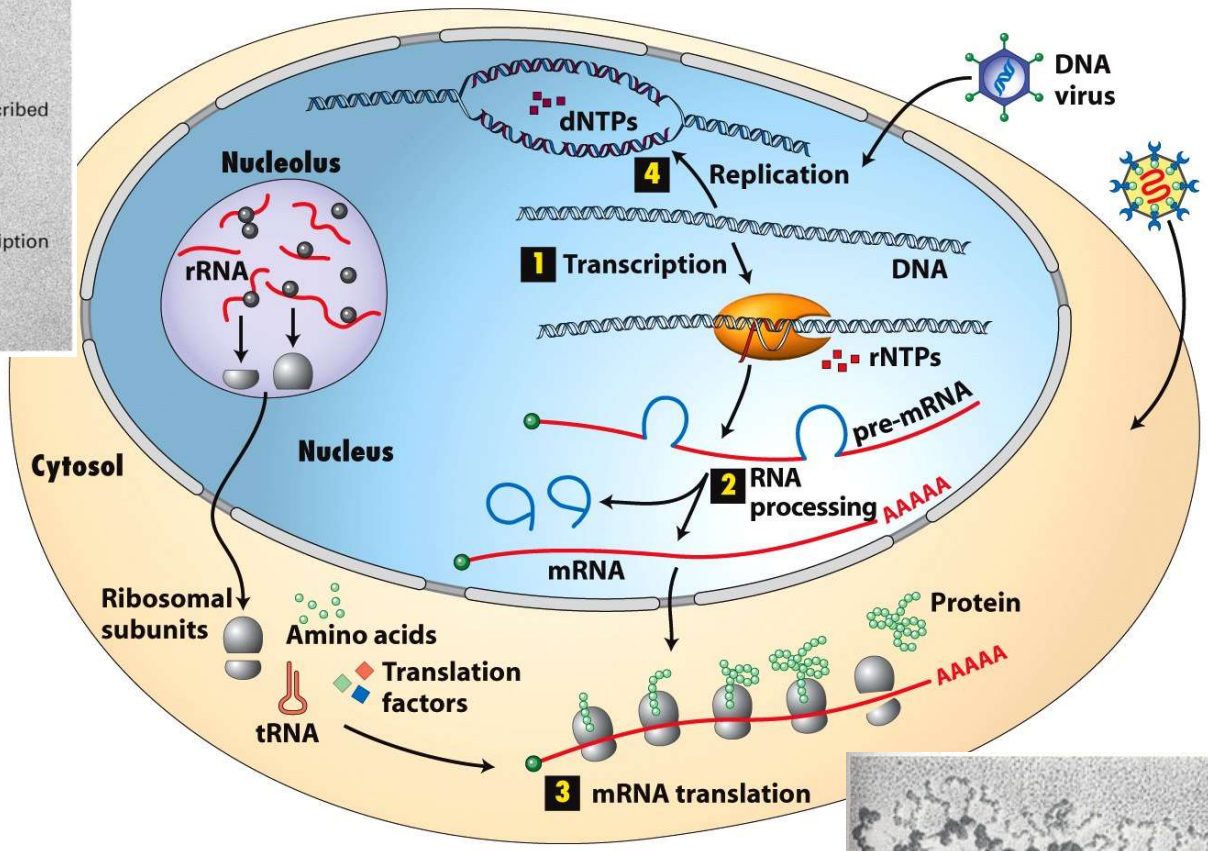
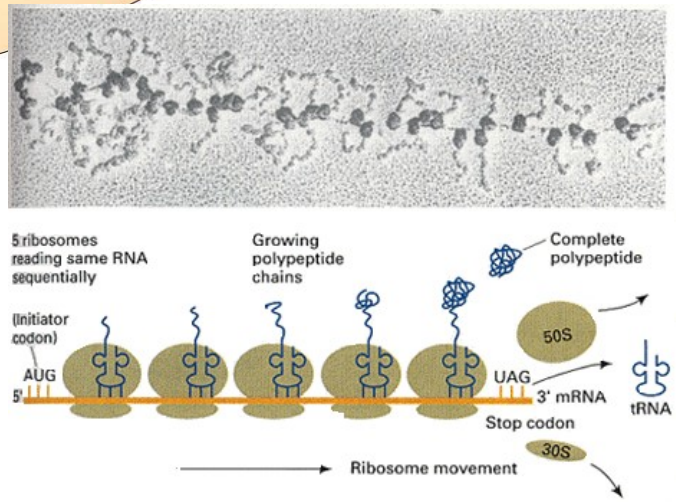
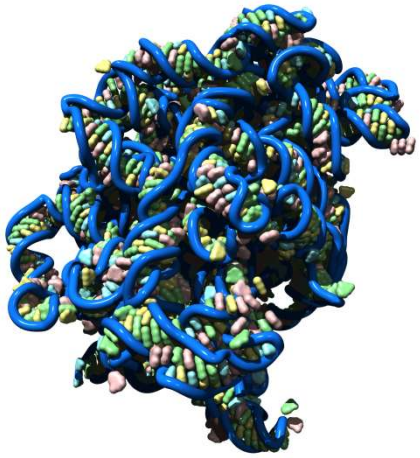


Figure 4-1
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A tradução do mRNA tem lugar no citoplasma



Ribossoma – estrutura macromolecular constituída por RNA ribossomal e proteínas



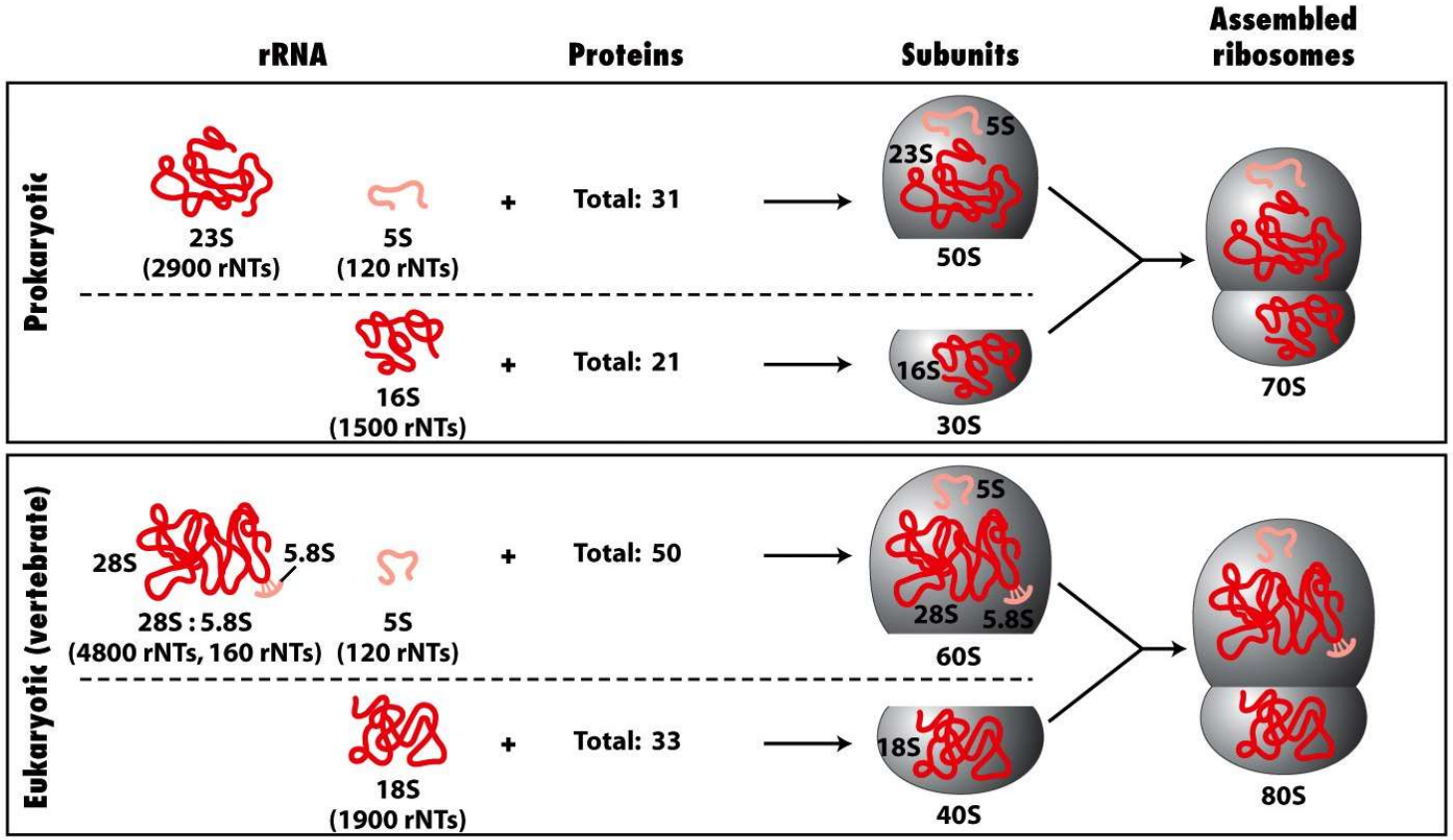
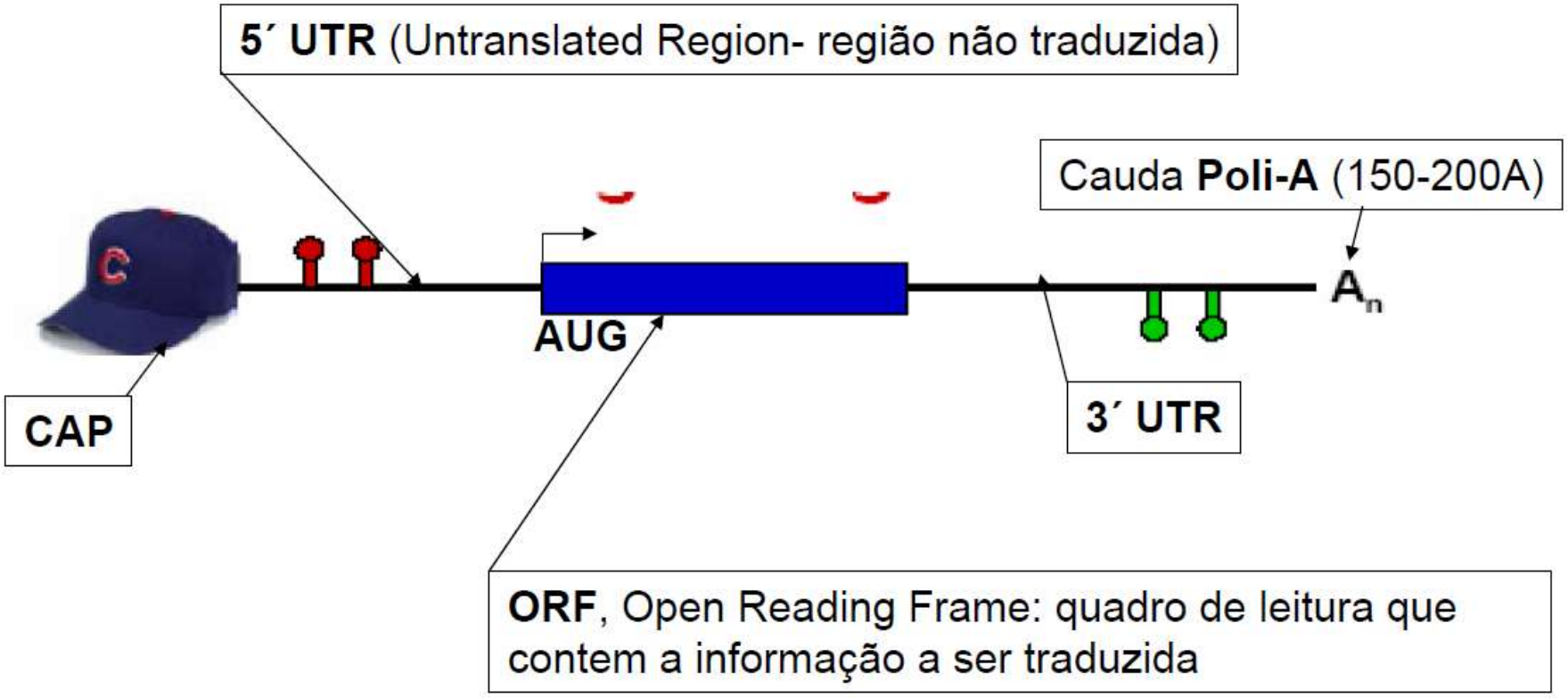
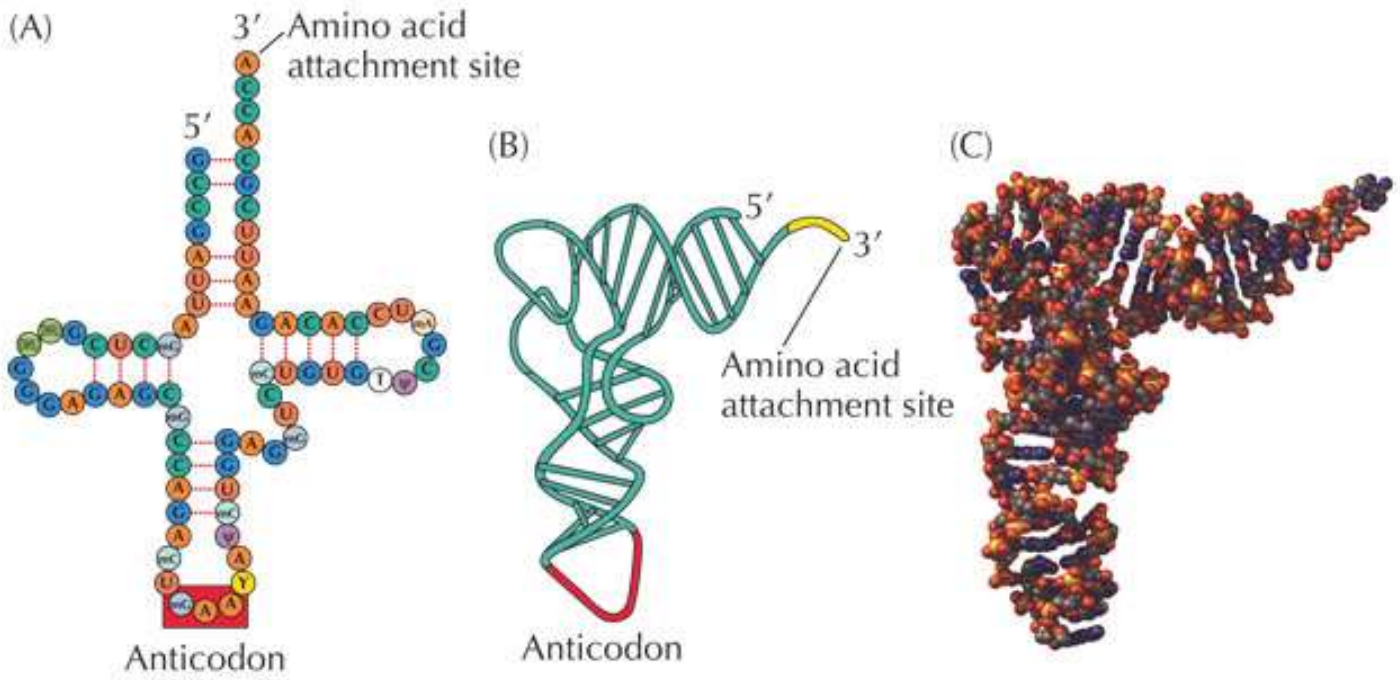


Figure 4-22
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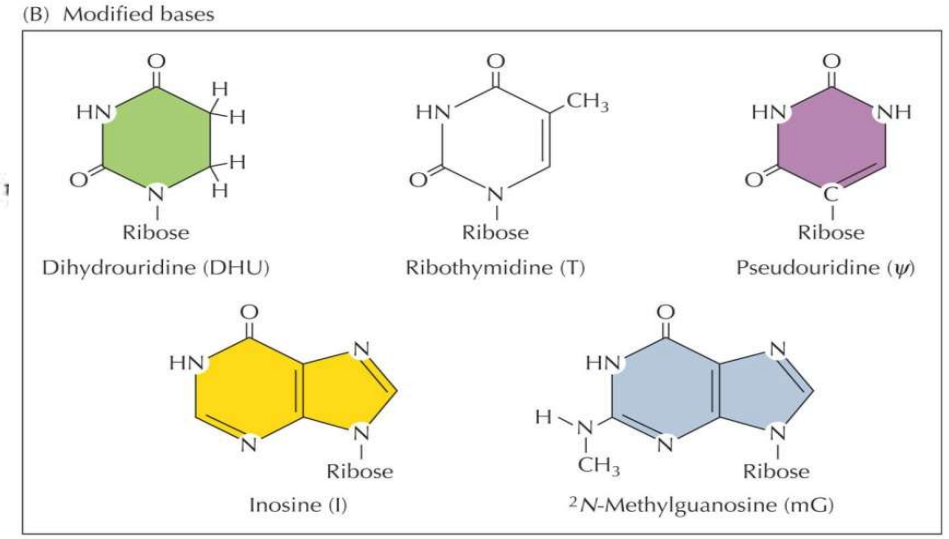
O mRNA nos eucariotas



RNA de transferência

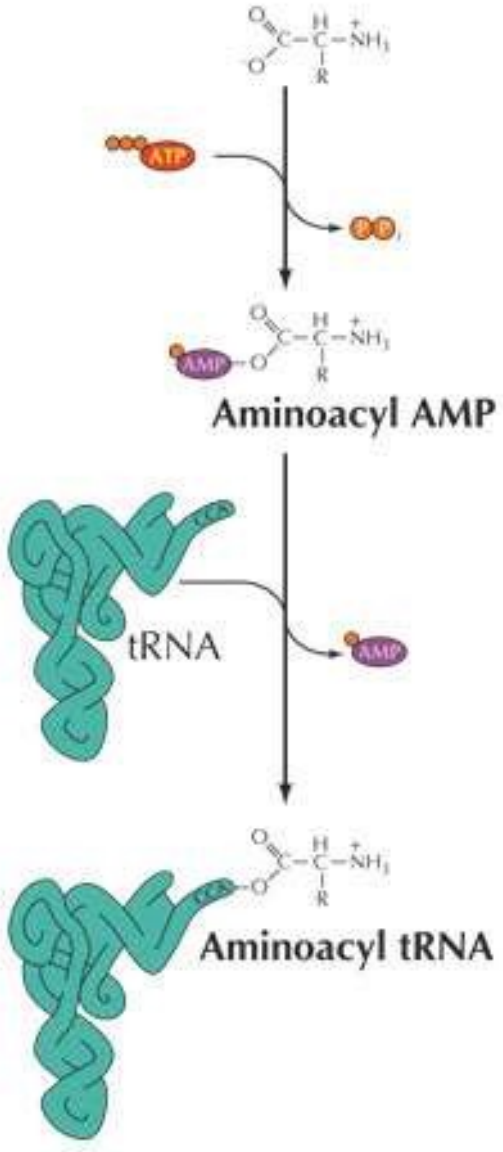


10% das bases nucleotídicas no tRNA são modificadas

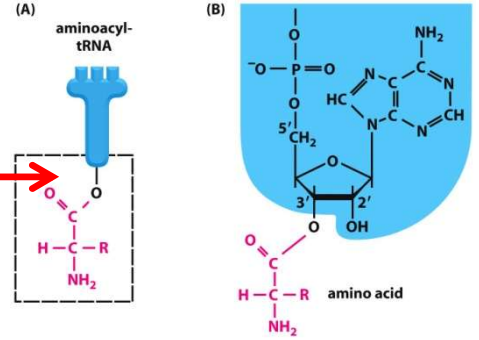


THE CELL, Third Edition, Figure 8.40 (Part 3) ASM Press and Sinauer Associates, Inc.
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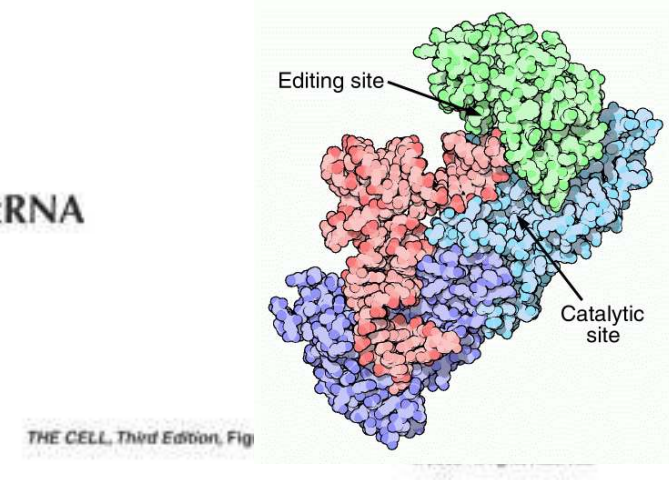
Ligação dos aminoácidos ao RNA de transferência



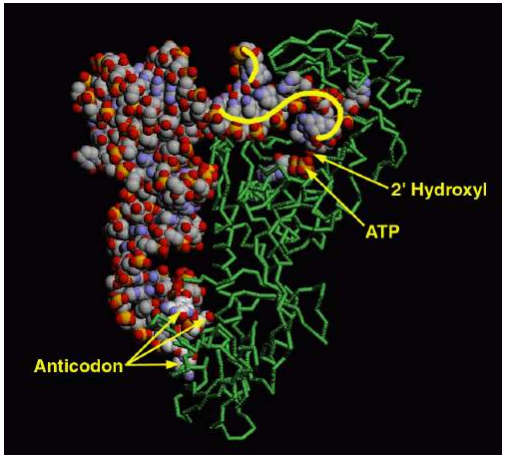
ligação éster com elevada energia



Há *proofreading* da reacção catalisada pelas aminoacil tRNA sintetases, baseado nas interacções entre o tRNA , o a.a. e o centro activo da enzima



THE CELL, Third Edition, Fig



Tradução do mRNA a proteínas

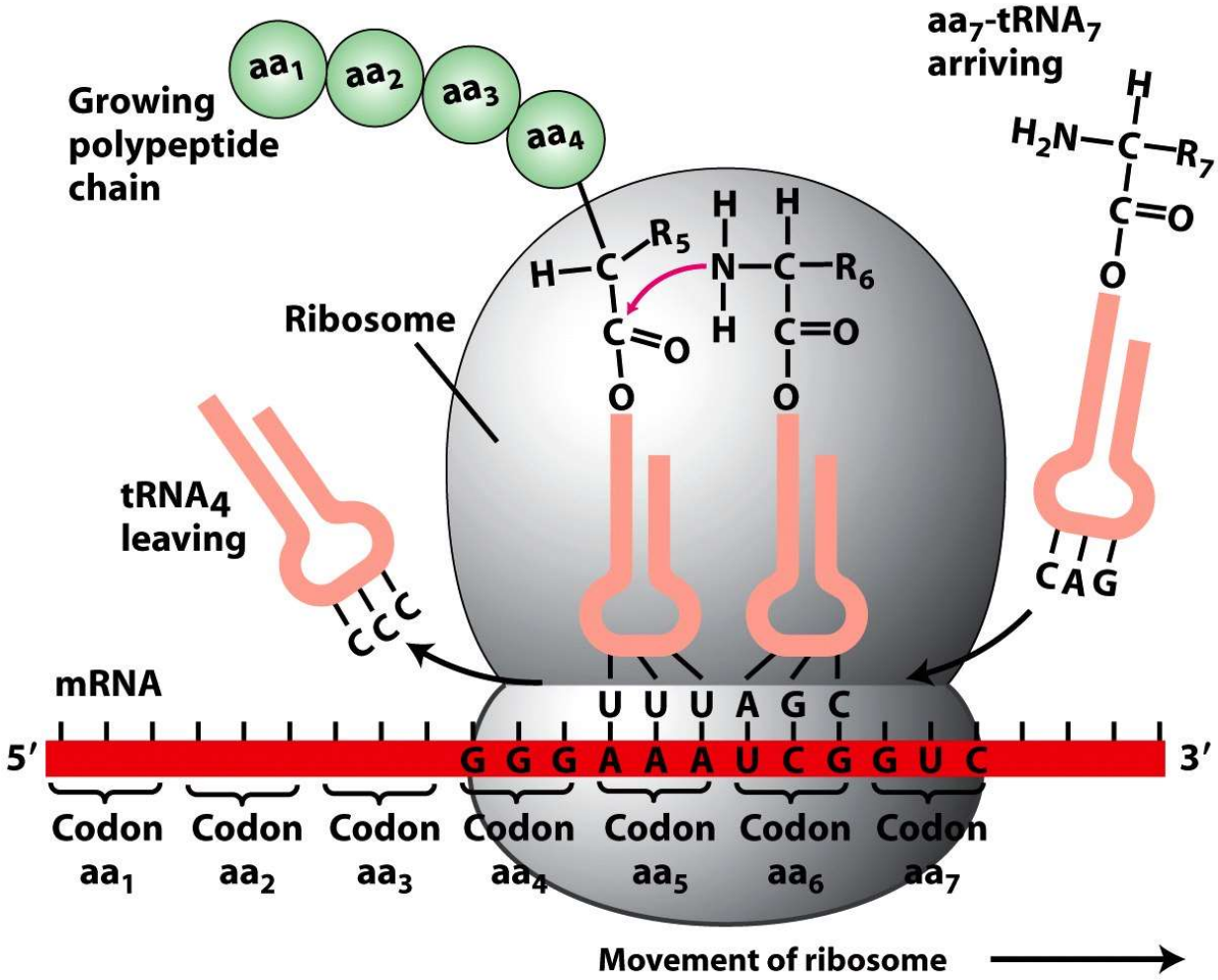




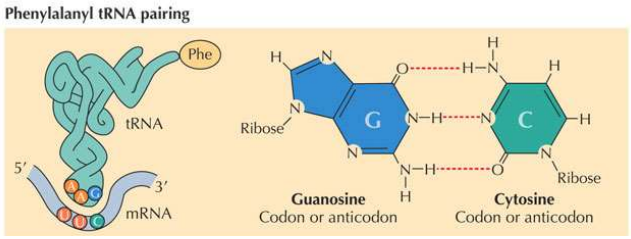
Figure 4-17
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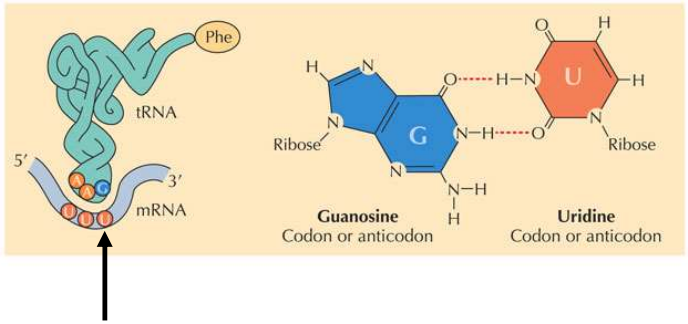
O código genético é degenerado

	U	C	A	G
U	UUU Phe UUC Phe UUA Leu UUG Leu	UCU Ser UCC Ser UCA Ser UCG Ser	UAU Tyr UAC Tyr UAA STOP UAG STOP	UGU Cys UGC Cys UGA STOP UGG Trp
C	CUU Leu CUC Leu CUA Leu CUG Leu	CCU Pro CCC Pro CCA Pro CCG Pro	CAU His CAC His CAA Gln CAG Gln	CGU Arg CGC Arg CGA Arg CGG Arg
A	AUU Ile AUC Ile AUA Ile AUG Met	ACU Thr ACC Thr ACA Thr ACG Thr	AAU Asn AAC Asn AAA Lys AAG Lys	AGU Ser AGC Ser AGA Arg AGG Arg
G	GUU Val GUC Val GUA Val GUG Val	GCU Ala GCC Ala GCA Ala GCG Ala	GAU Asp GAC Asp GAA Glu GAG Glu	GGU Gly GGC Gly GGA Gly GGG Gly

Cada aa é especificado por um ou mais tripletos da bases- codão – **61 codões** codificantes de aa

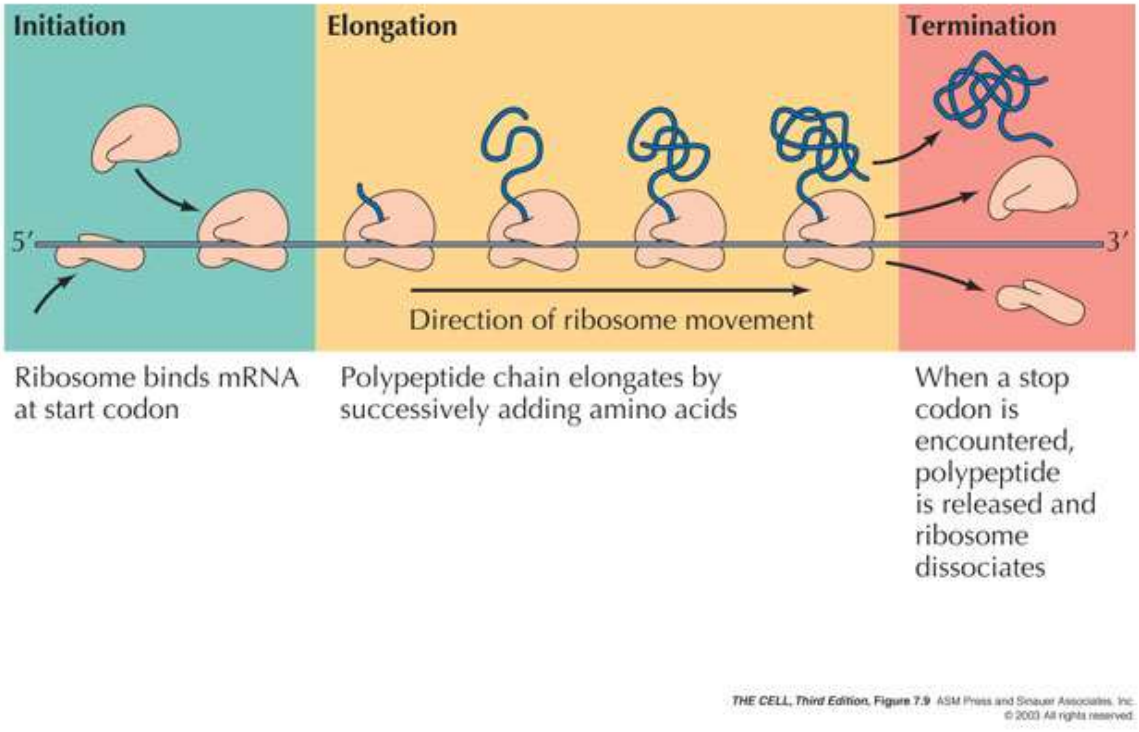
Existe emparelhamento não-standard entre codão e anti-codão – **48 sequências de anti-codão** na espécie humana - **20 aa**

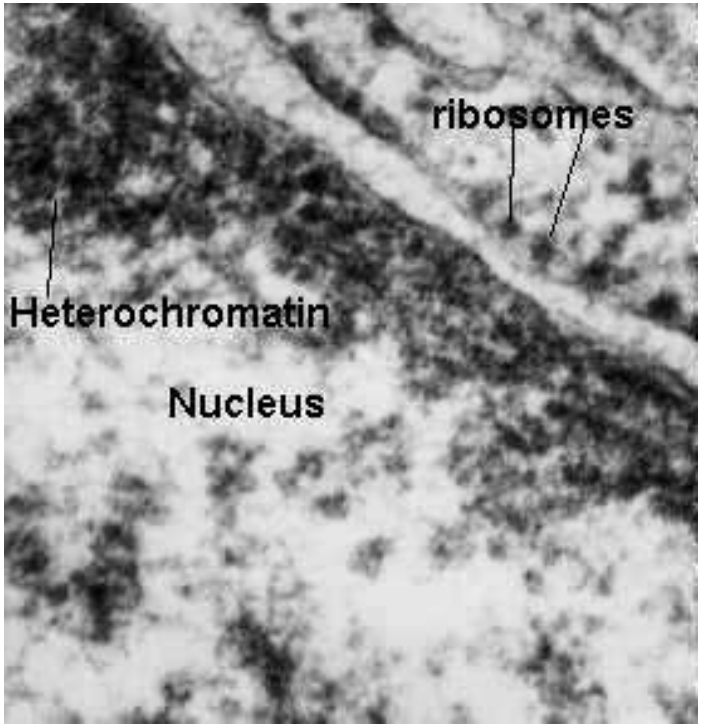




Posição vacilante (*wobble*)

Iniciação, elongação e terminação da tradução





Iniciação da tradução

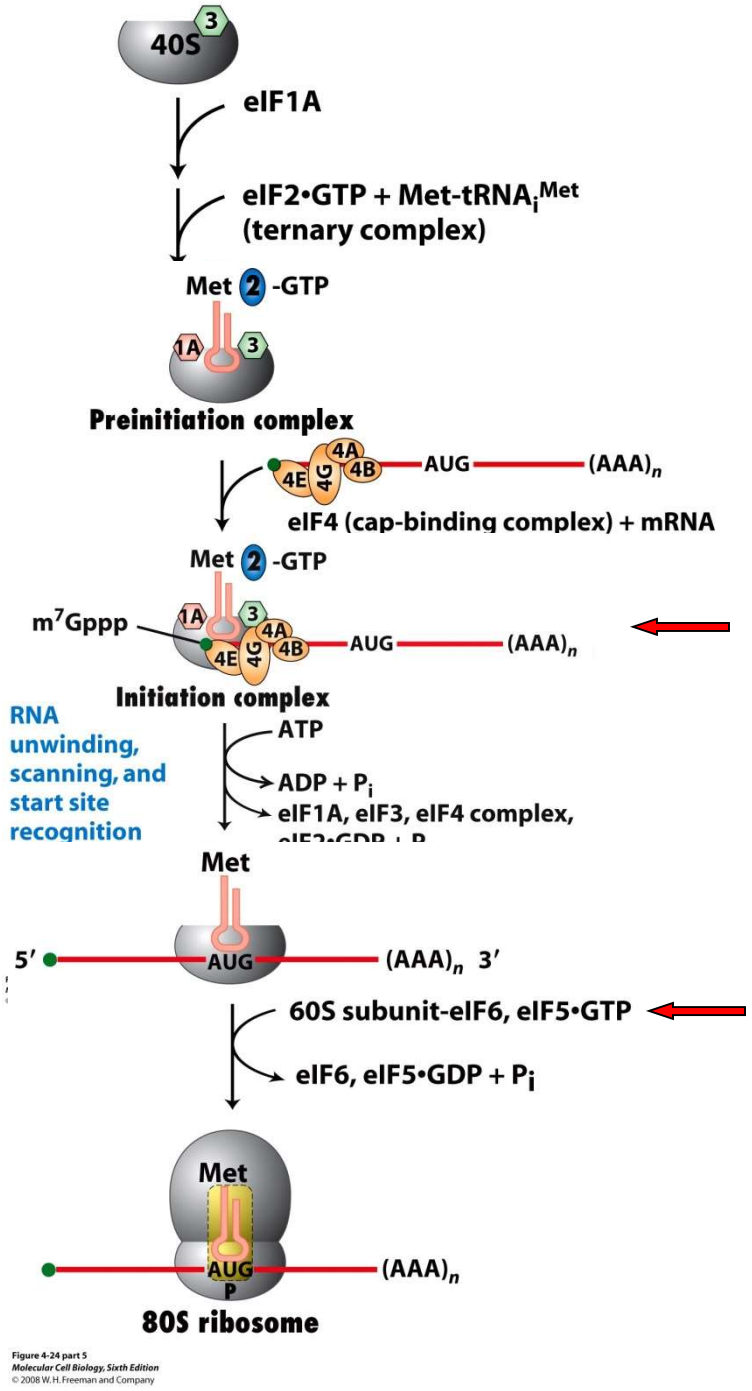
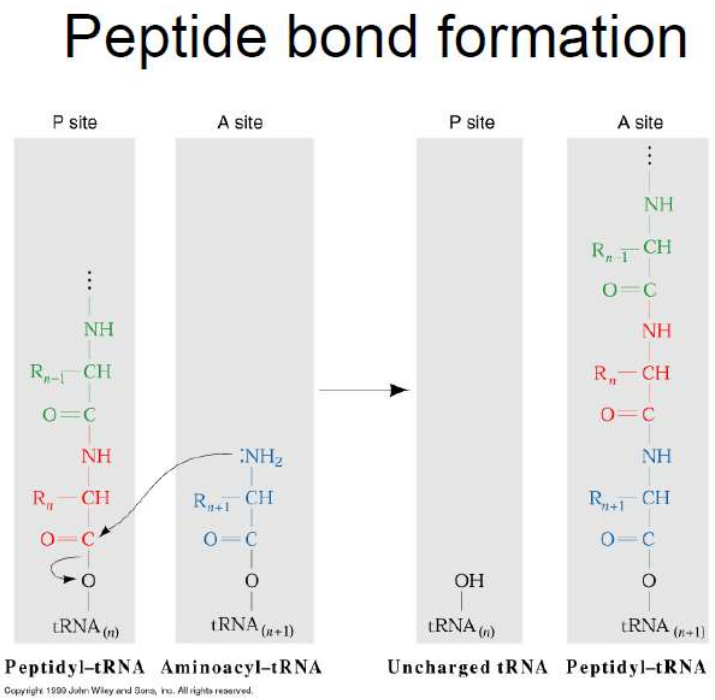
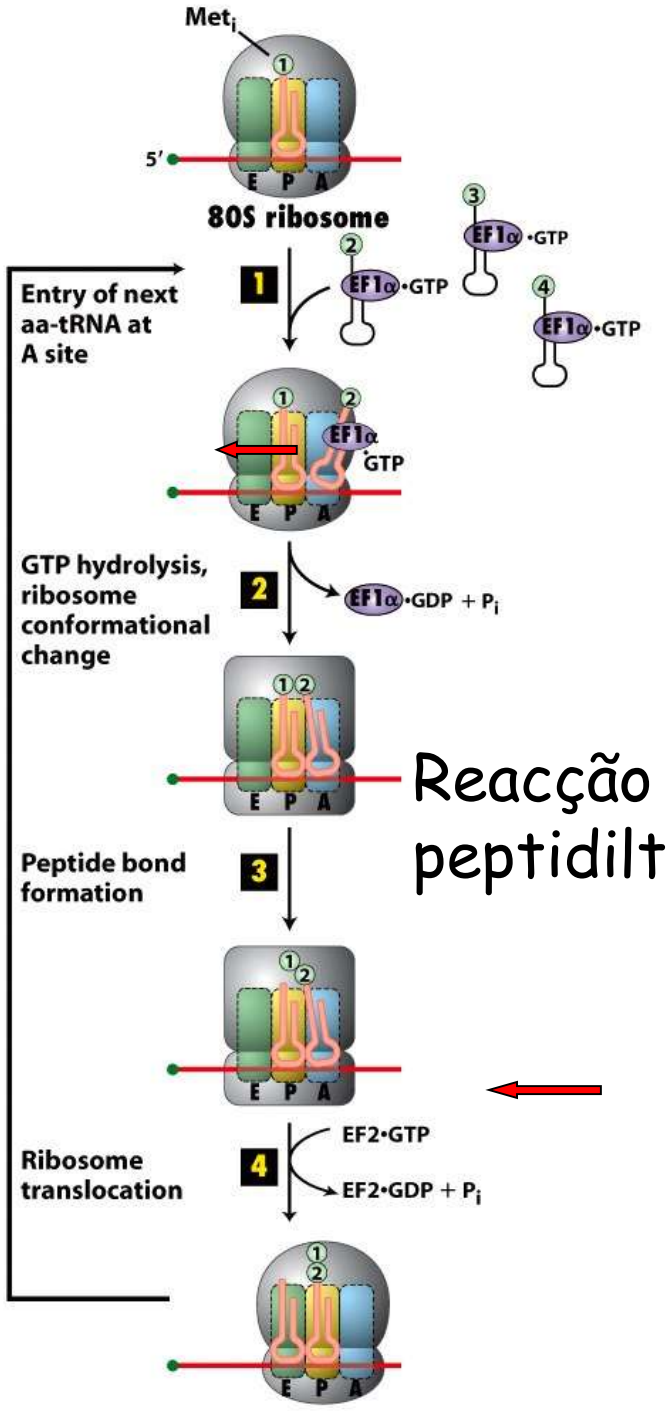


Figure 4-24 part 5
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Elongação



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Reacção de peptidiltransferase

Figure 4-25

Terminação da tradução

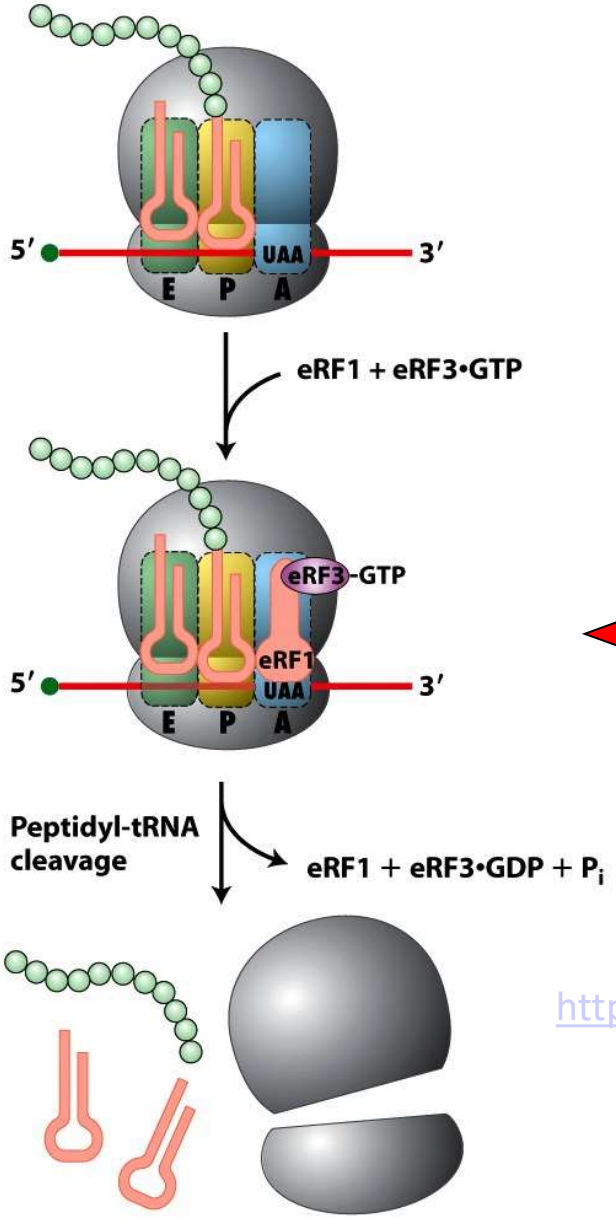
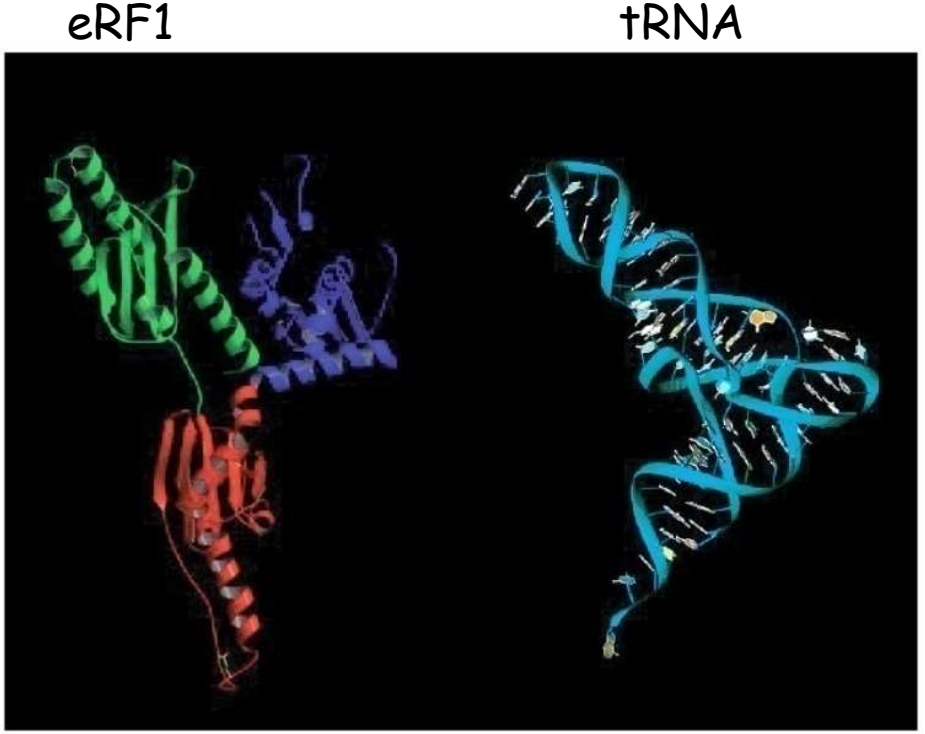


Figure 4-27
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http://www.youtube.com/watch?v=5bLEDd-PSTQ&feature=related

Bibliografia:

Lodish et al. 9 ª ed. “Molecular Cell Biology” Freeman 2021

Alberts 7ª ed. “Molecular Biology of the Cell” Garland 2022